



Data Mining Perspective On RAG : Research & Practice

2026-02-05

Presenter : Sooho Moon

MINDS Lab

INDEX

■ REFERENCING FROM OUTER SOURCE

■ RECENT METHODS

■ RESEARCH TOPICS

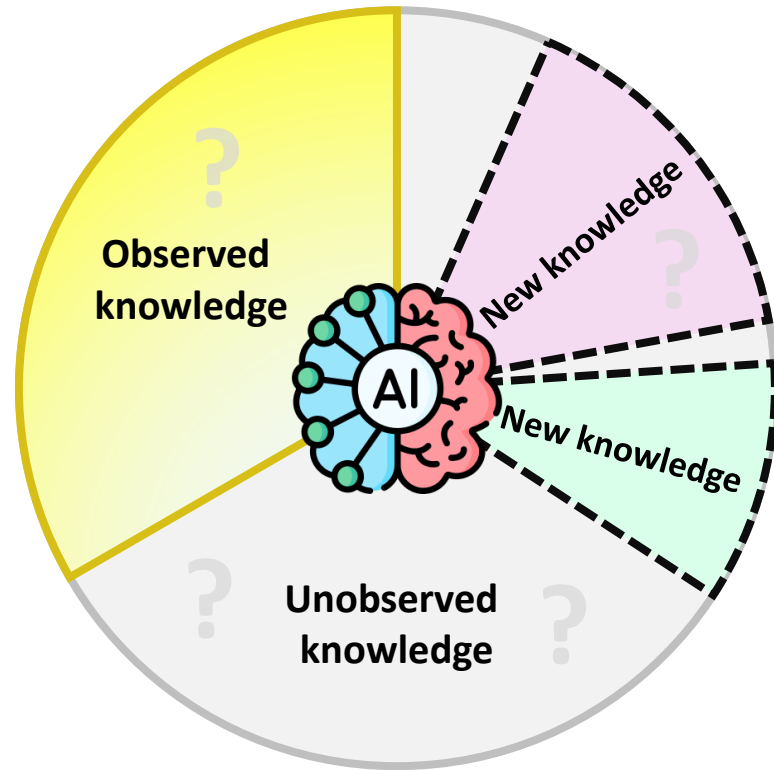
■ DOCUMENT RETRIEVAL (WSDM CUP 26)

■ EXPERIMENTS & DISCUSSION

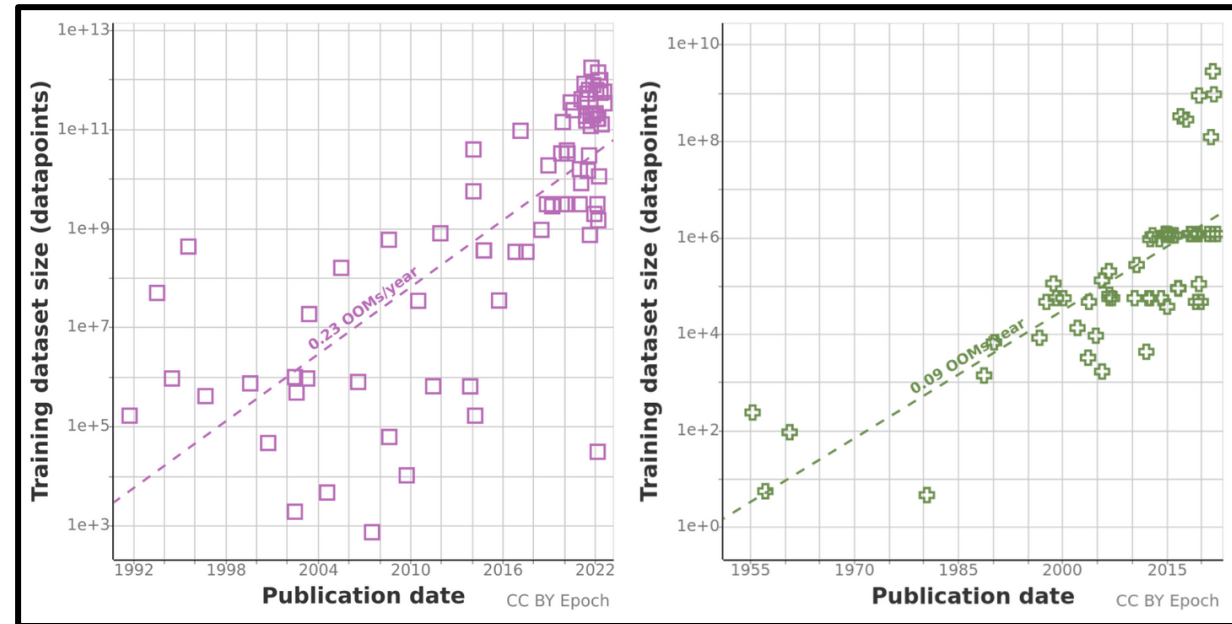
■ CONCLUSION

REFERENCING FROM OUTER SOURCE

■ Costly Training & Forgetting

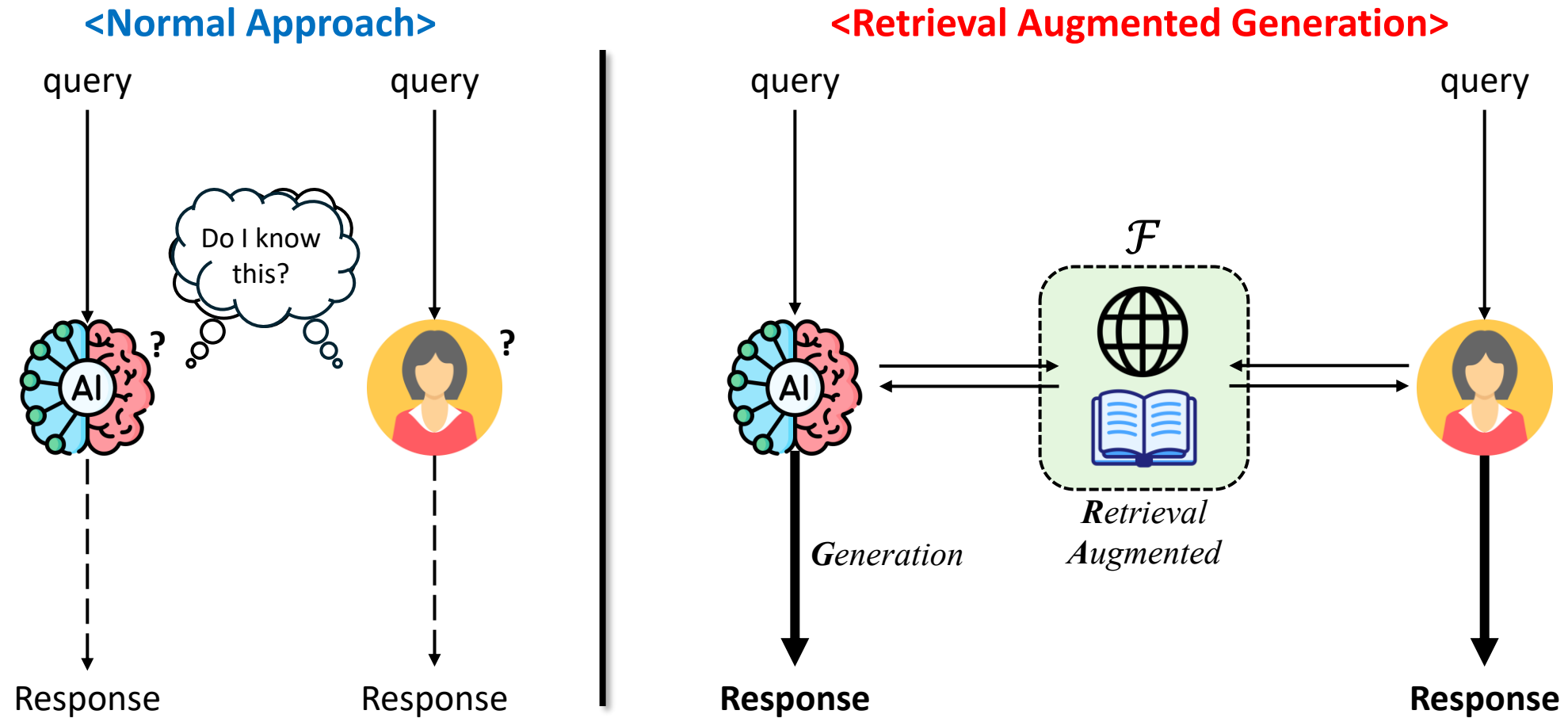


Exponential growth of data size (NLP & Vision)



REFERENCING FROM OUTER SOURCE

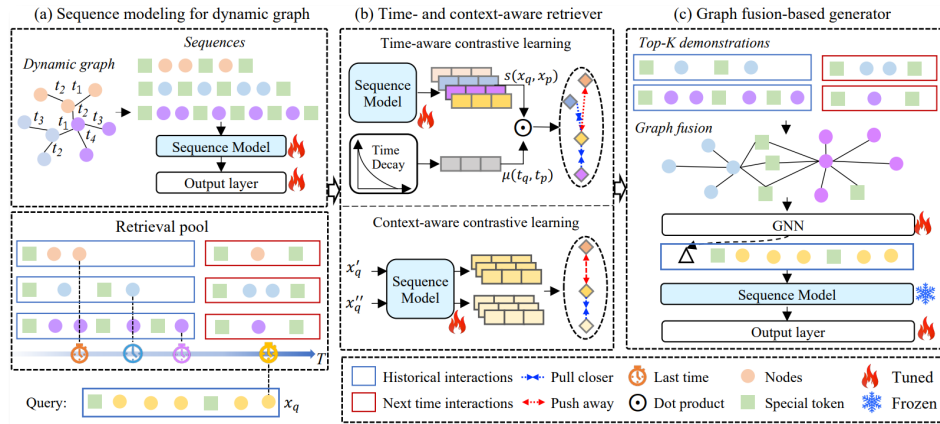
■ Providing Reference For Accurate & Reliable Response



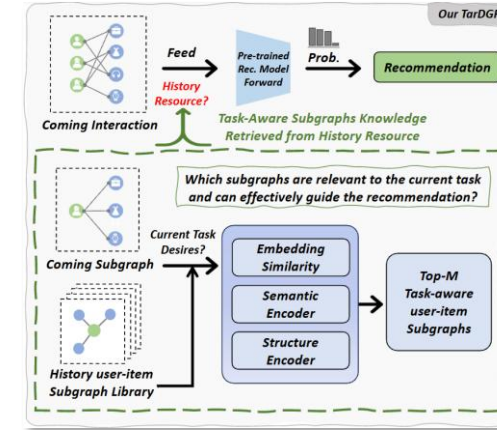
REFERENCING FROM OUTER SOURCE

Recent Adaptations

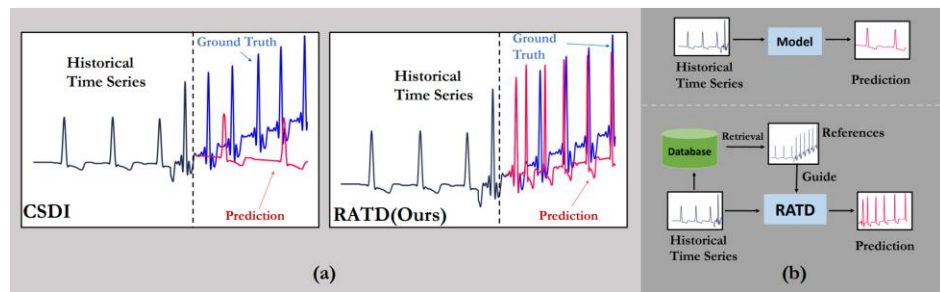
RAG for Dynamic Graph Learning (SIGIR'25)



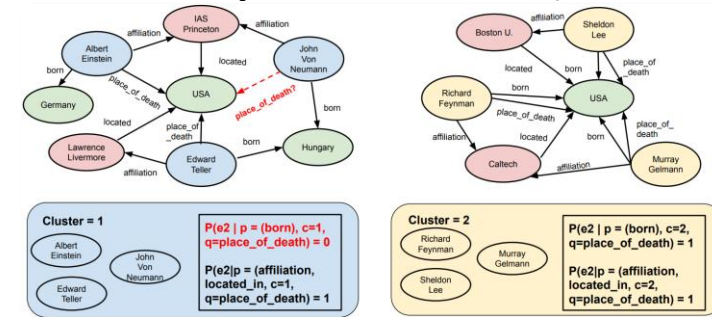
RAG for Dynamic Recommendation (AAAI'26)



Reference Guided TS Forecasting (NeurIPS'24)



Case-based Open World KGC (EMNLP'20)



Wu, Yuxia, Lizi Liao, and Yuan Fang. "Retrieval augmented generation for dynamic graph modeling." Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval. 2025.

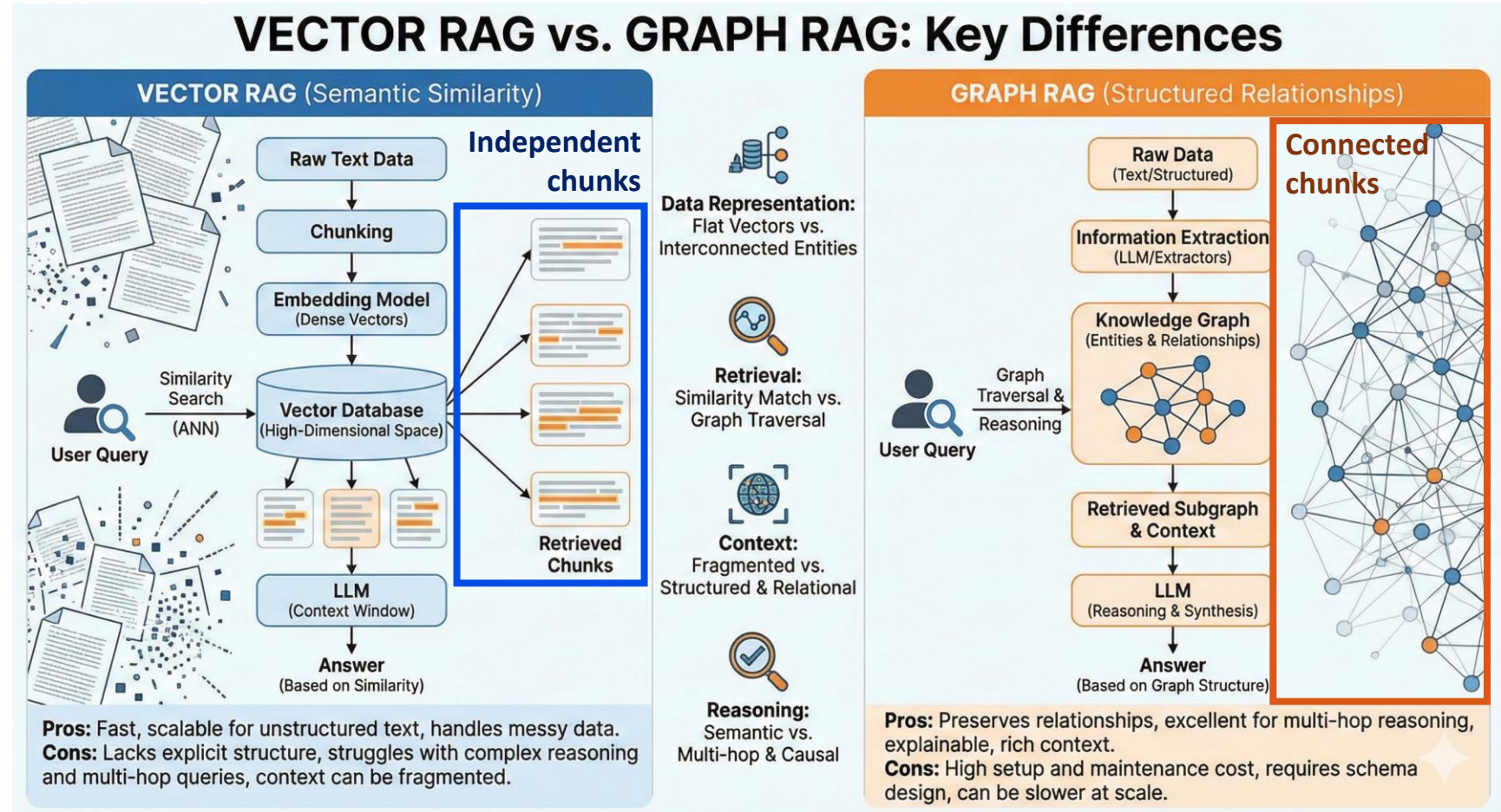
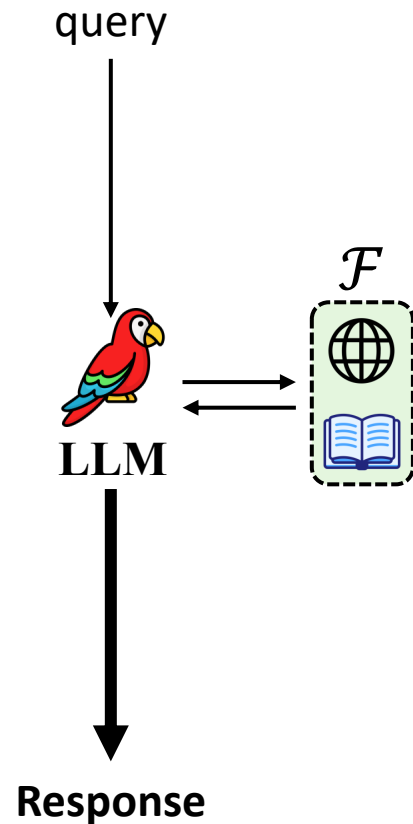
Tao, Zhen, et al. "Task-Aware Retrieval Augmentation for Dynamic Recommendation." *arXiv preprint arXiv:2511.12495* (2025).

Liu, Jingwei, et al. "Retrieval-augmented diffusion models for time series forecasting." *Advances in Neural Information Processing Systems* 37 (2024): 2766-2786.

Das, Rajarshi, et al. "Probabilistic Case-based Reasoning for Open-World Knowledge Graph Completion." *Findings of the Association for Computational Linguistics: EMNLP 2020*. 2020.

REFERENCING FROM OUTER SOURCE

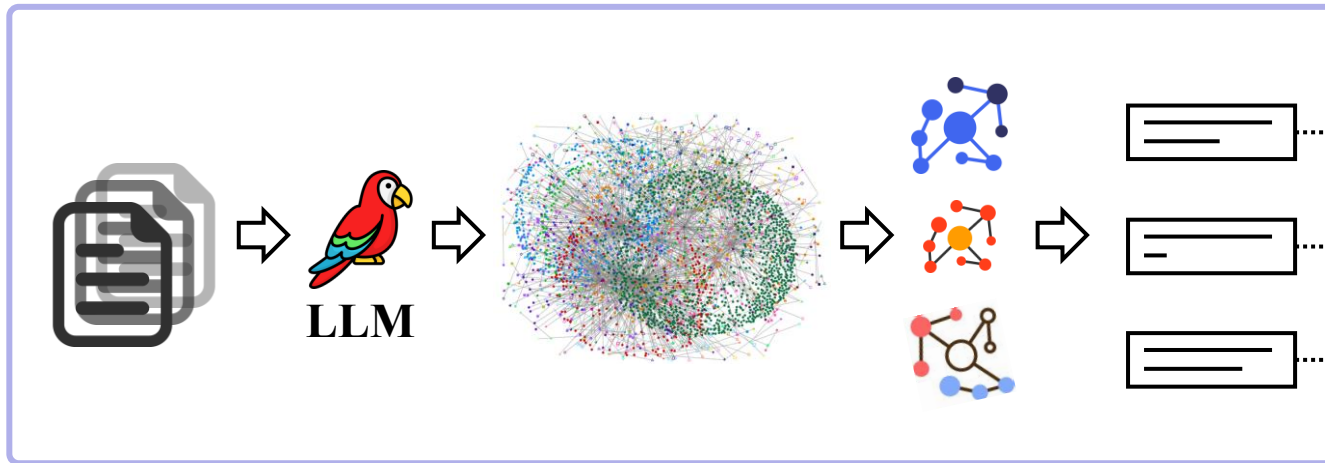
■ LLM View : Formulizing Reference Pool (VectorRAG, GraphRAG)



RECENT METHODS

■ GraphRAG (arXiv 24)

Indexing Time

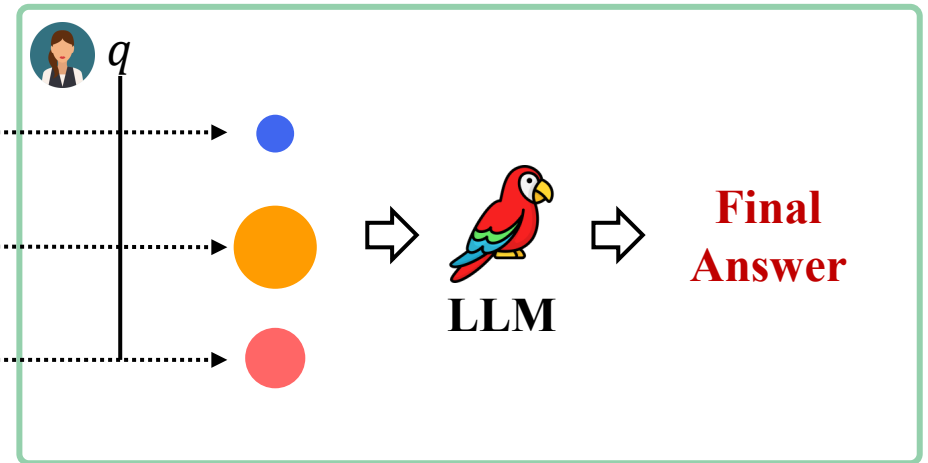


KG Construction

*Community
Detection*

Summarization

Retrieval & Query Time



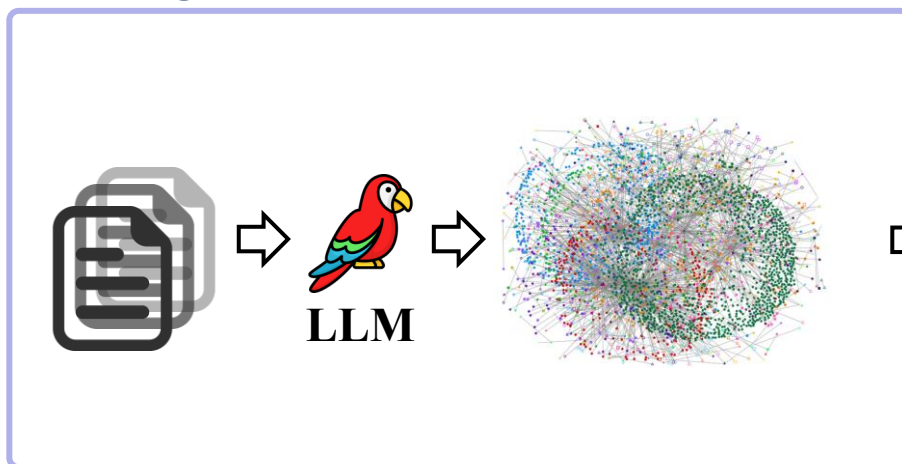
*Intermediate
Answers*

**Final
Answer**

RECENT METHODS

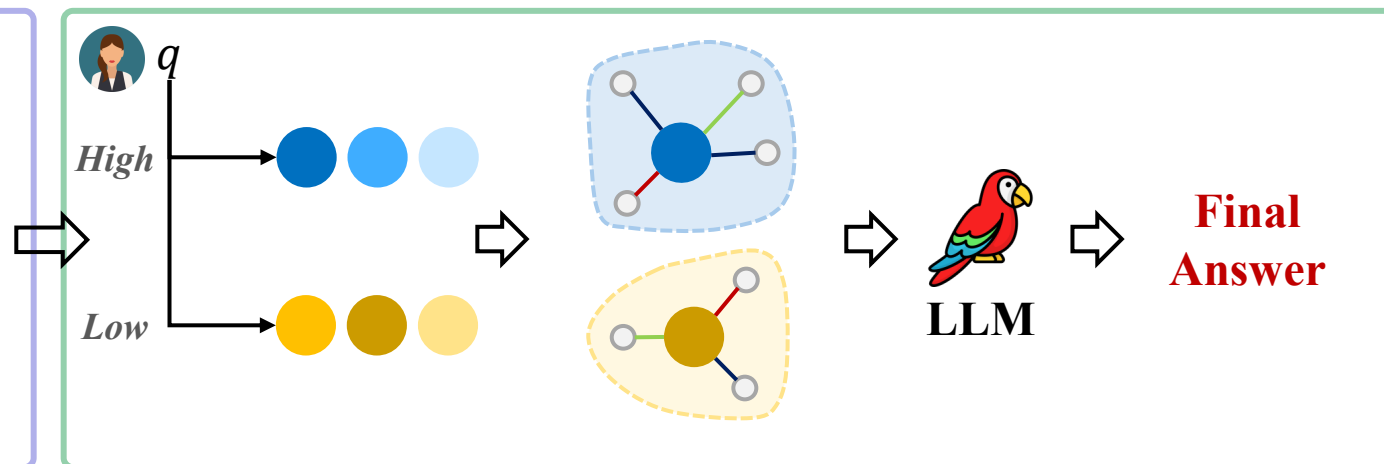
■ LightRAG (arXiv 24)

Indexing Time



KG Construction

Retrieval & Query Time



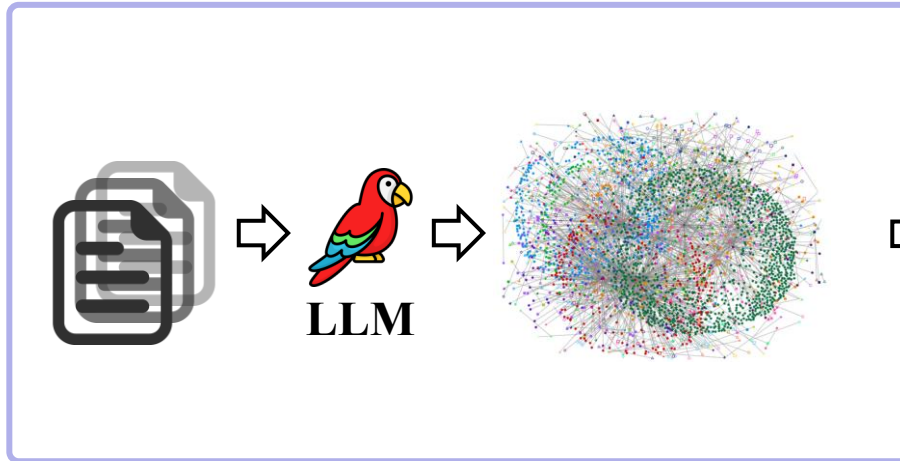
Dual Level Extraction

Neighbor Context

RECENT METHODS

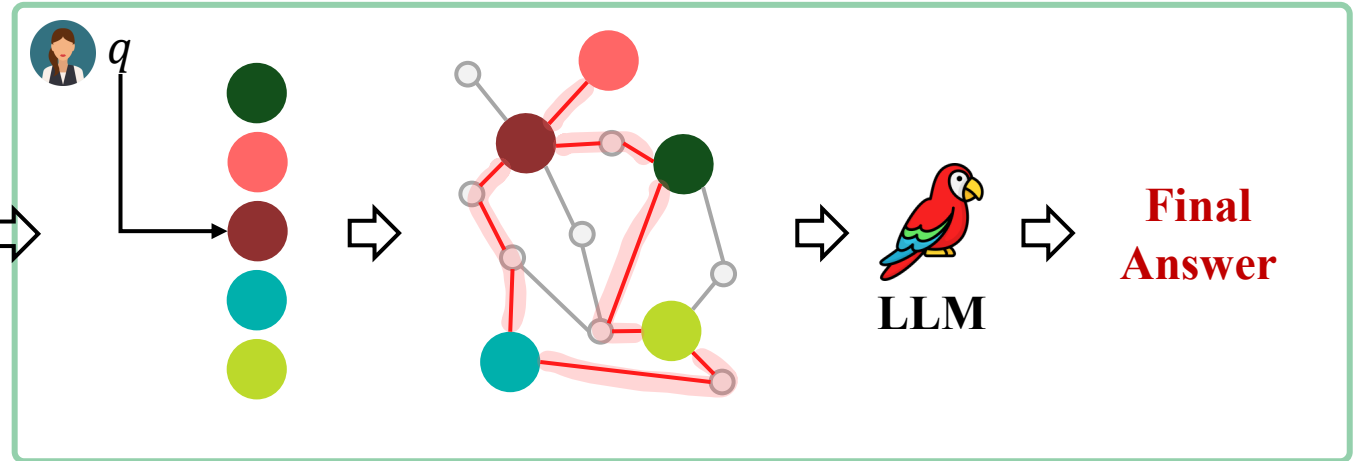
■ PathRAG (AAAI 25)

Indexing Time



KG Construction

Retrieval & Query Time



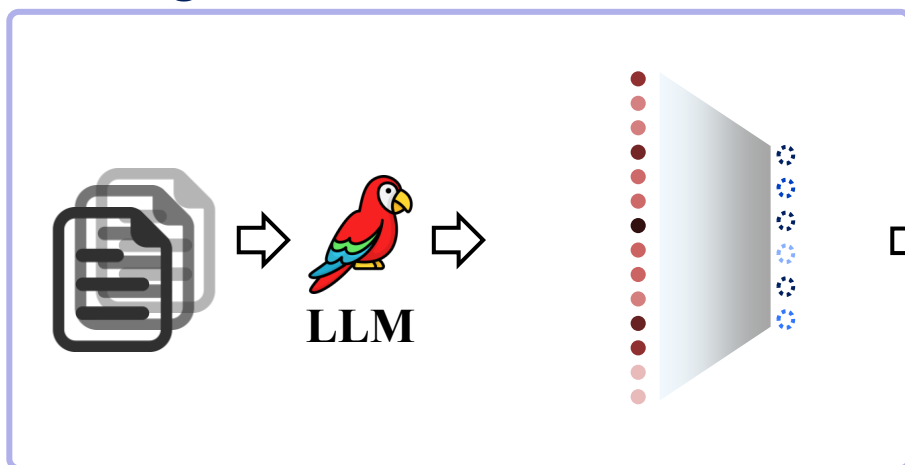
Keyword Extraction

Path Extraction

RECENT METHODS

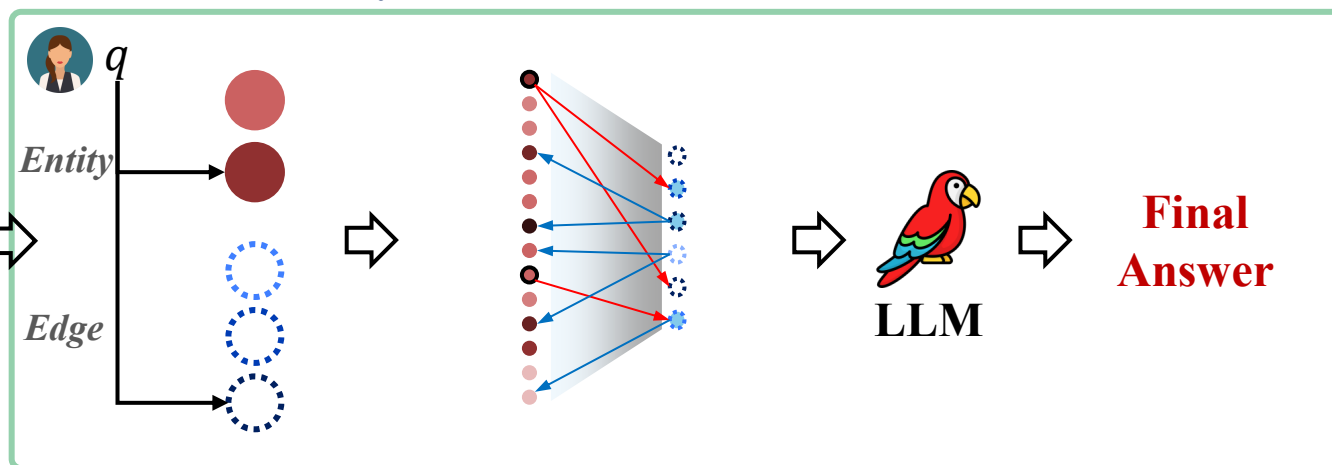
■ HyperGraphRAG (NeurIPS 25)

Indexing Time



Hyper KG Construction

Retrieval & Query Time

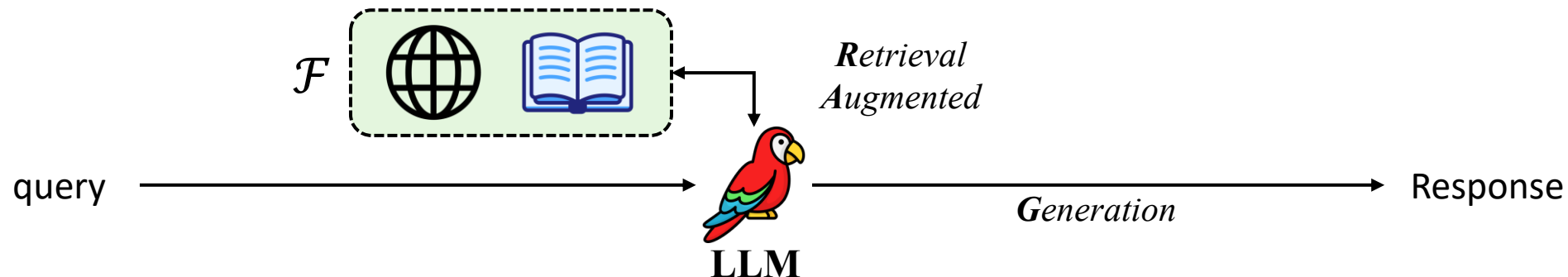
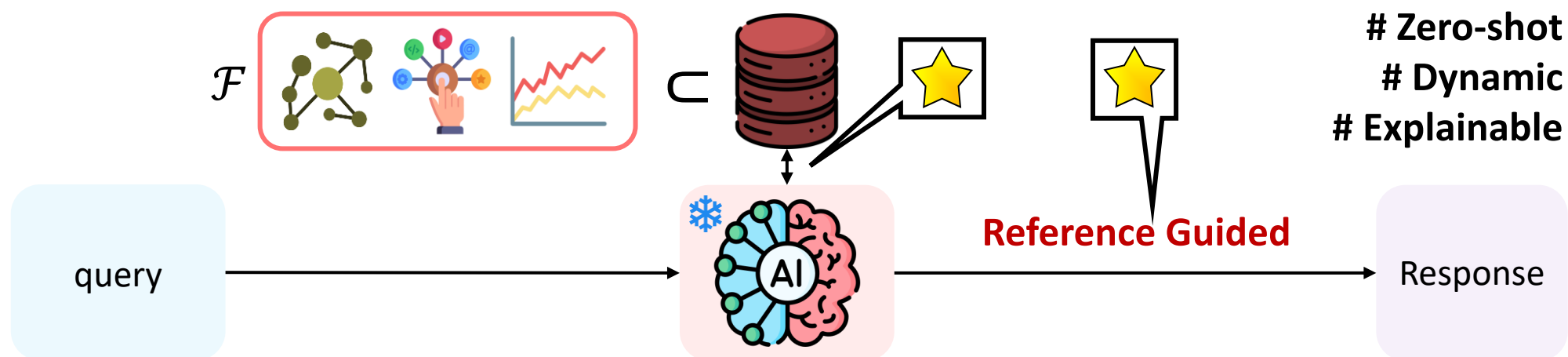


Keyword Extraction

Knowledge Fusion

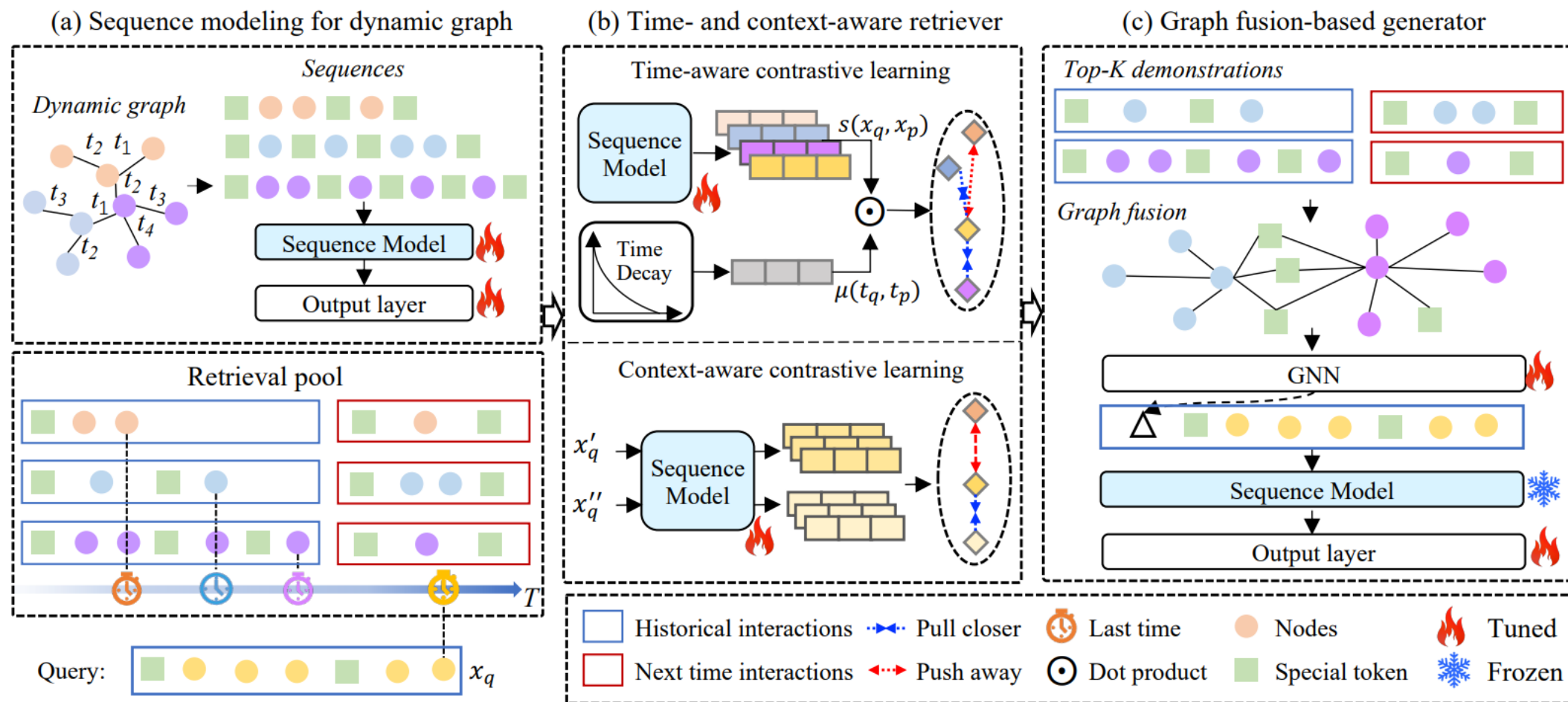
RESEARCH TOPICS

■ More Than Just a LLM Sub-Module



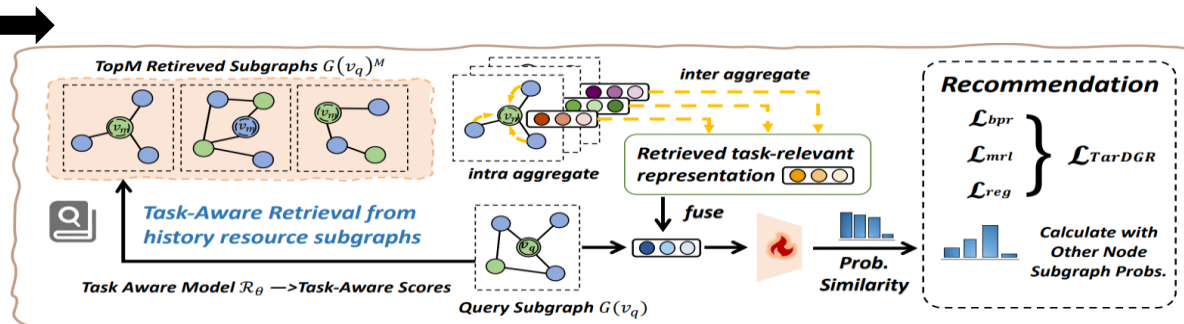
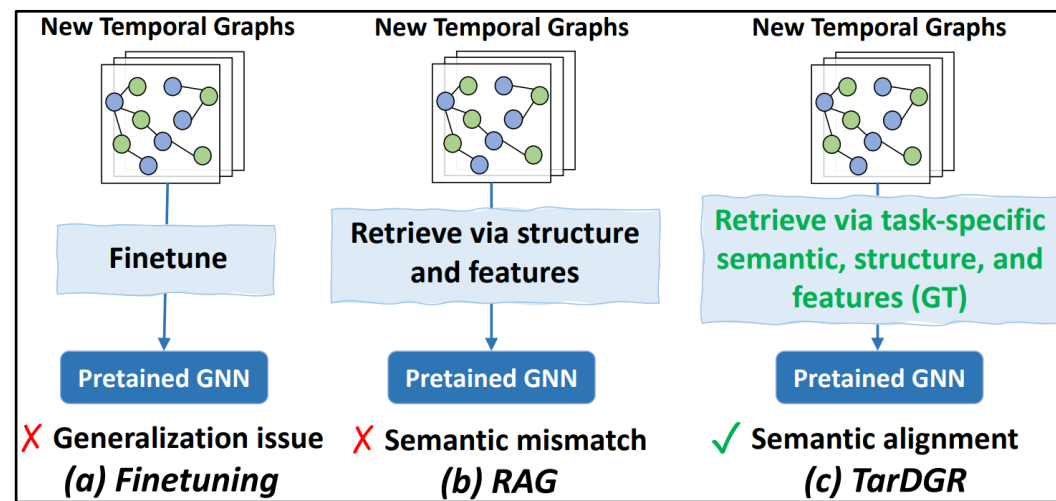
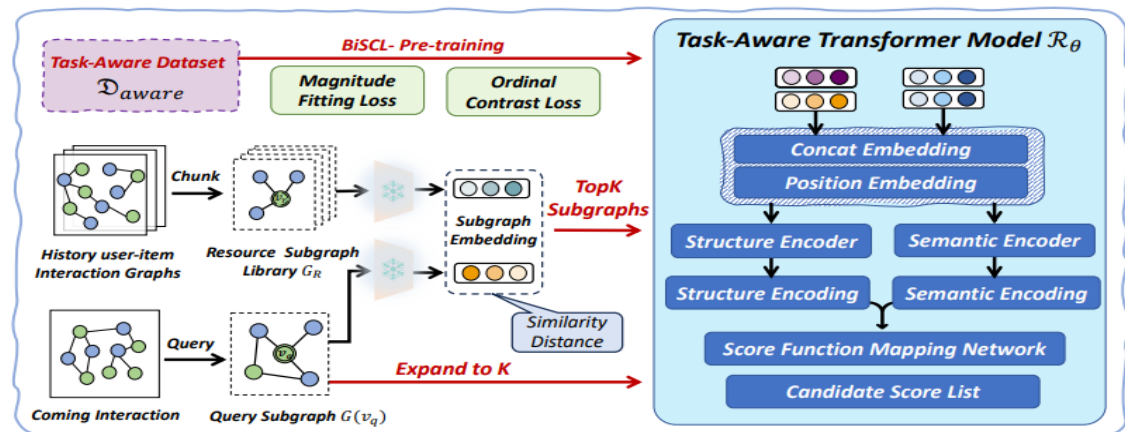
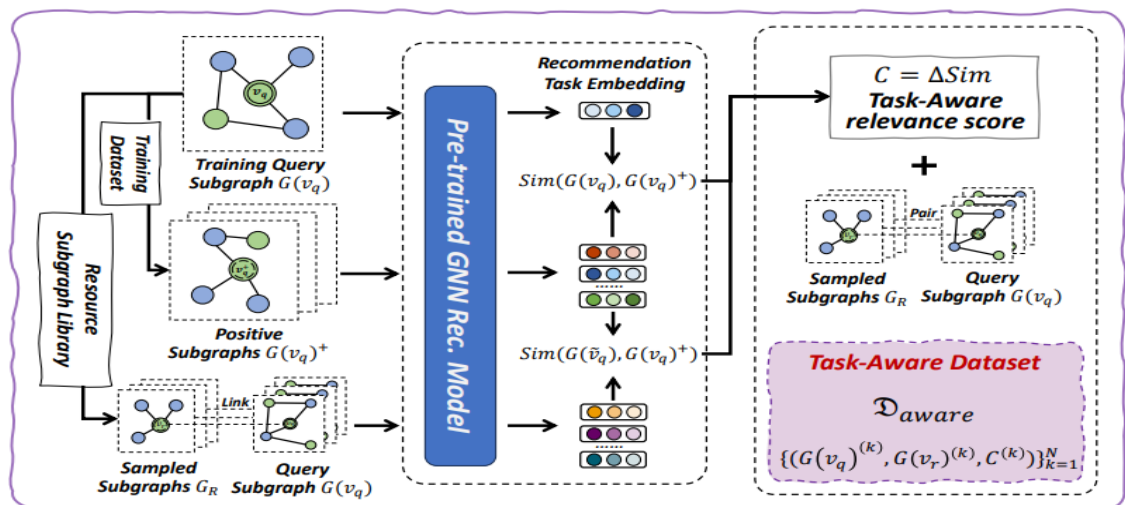
RESEARCH TOPICS

■ T1 : Retrieval Augmented Dynamic Graph Link Prediction (SIGIR'25)



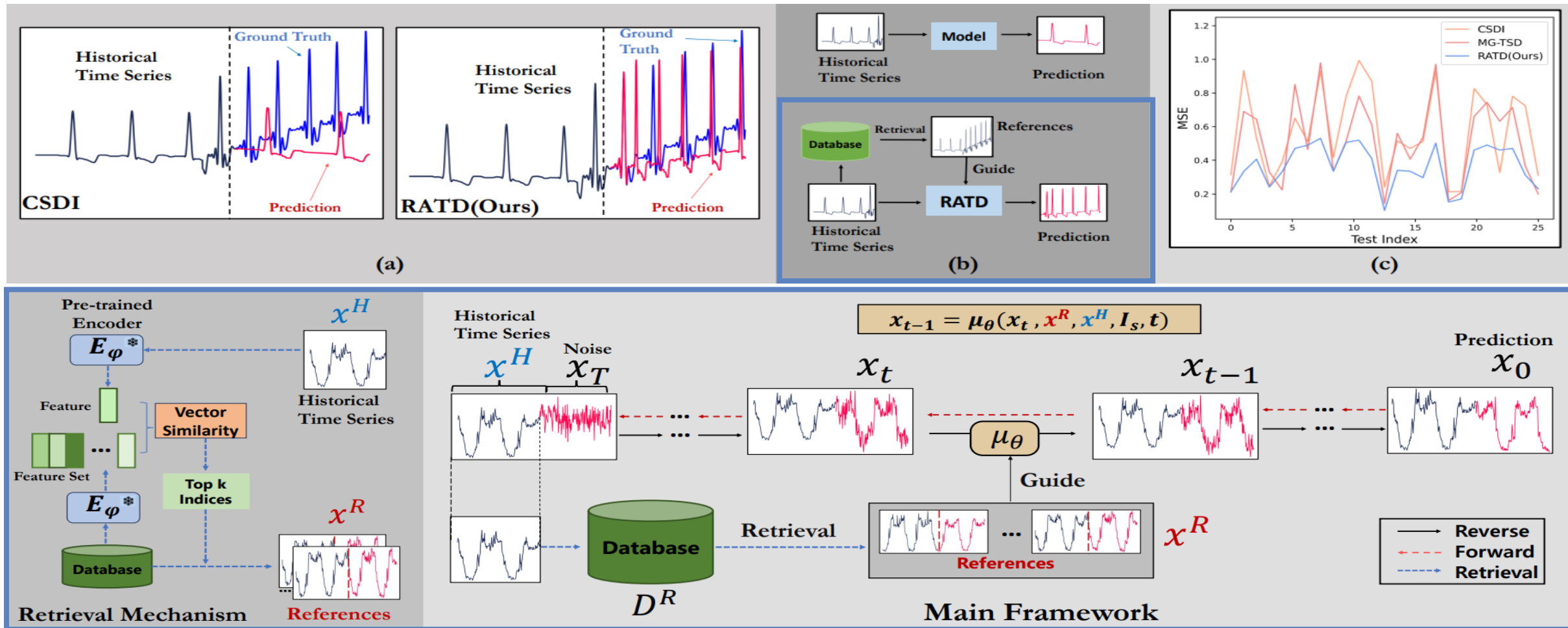
RESEARCH TOPICS

■ T2 : Retrieval Augmentation for Dynamic Recommendation (AAAI'26)



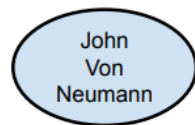
RESEARCH TOPICS

■ T3 : Reference Guided TS Forecasting (NeurIPS'24)



■ T4 : Case-based Open-World Knowledge Graph Completion (EMNLP'20)

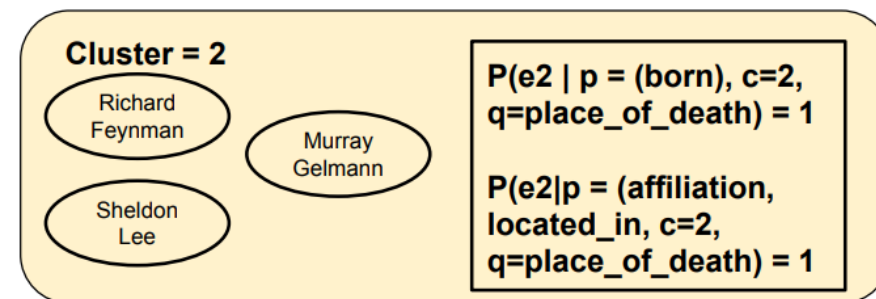
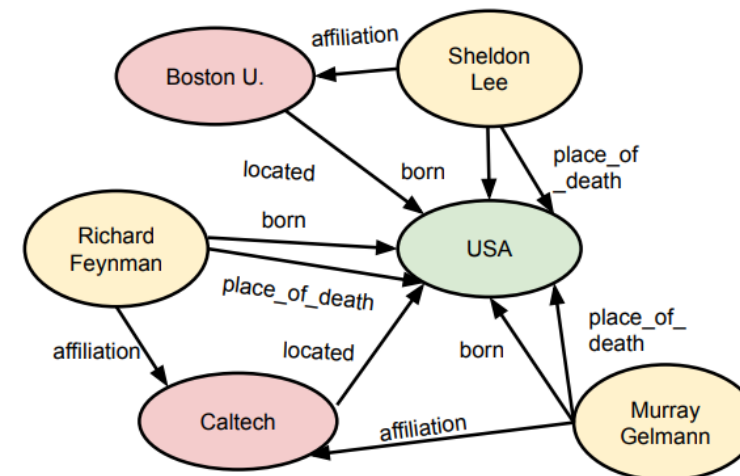
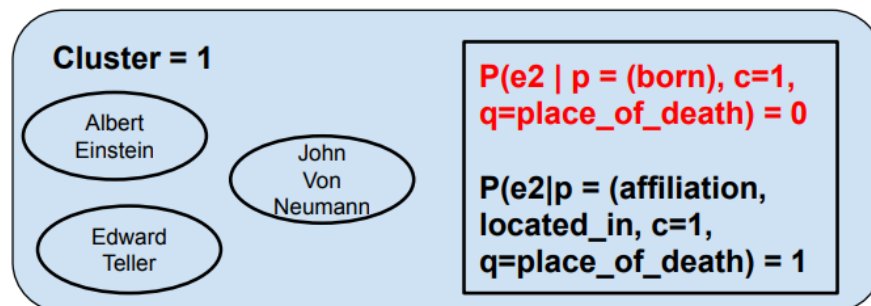
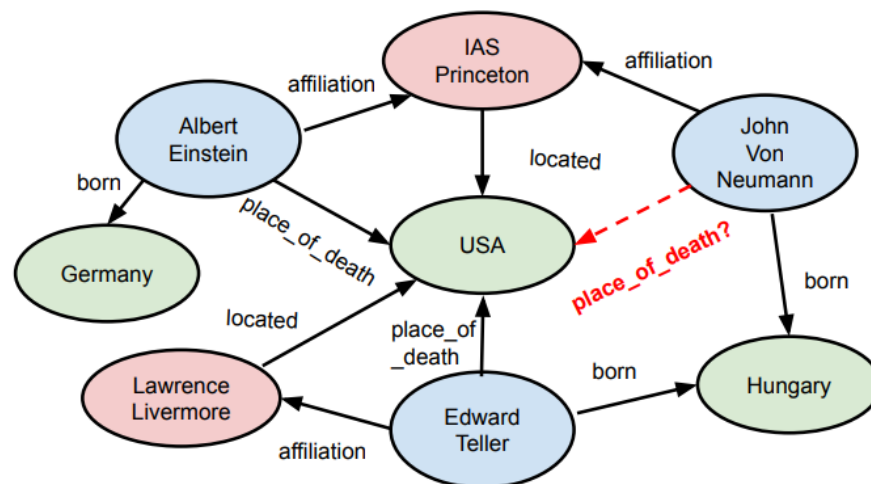
Query: (Jon Von Neumann, place_of_death, ?)



KNN-Search

Albert Einstein, Edward Teller, Richard Feynman, Murray Gelmann, Sheldon Lee.

Paths gathered from KNN entities:
1) $place_of_death(x,y) \Leftarrow born(x,y)$
2) $place_of_death(x,y) \Leftarrow affiliation(x,z) \wedge located(z,y)$

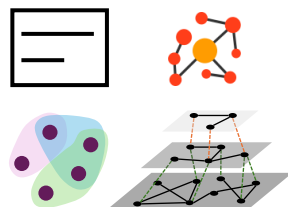


■ Key Challenges For Future Retrieval Augmentation & Generation

Reference Pool

What form?

Text chunks , KGs, KHGs,
Hierarchical Graphs, ~



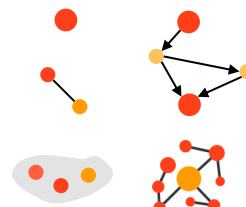
How?

LLMs, Train samples ~

Retrieval Method

Which Granularity?

Nodes, Triplets, Paths,
Hyperedges , Subgraphs, ~



How?

Non-Parametric, LLMs
Random walks, GNNs ~

Q&A

What Query?

Text, Node similarity, ~



Query Processing?

Raw, Decomposition, ~

How?

LLMs, Model output, ~

Is it efficient?

Time, Memory, ~

DOCUMENT RETRIEVAL (WSDM CUP 26)

■ Multilingual Document Retrieval

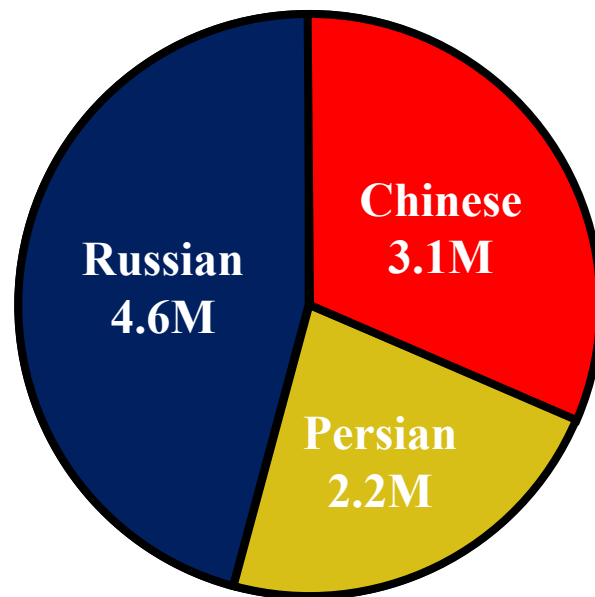
(1) English Query



Query

*"I am looking for foods
that may cause anger
issues in humans."*

(2) Document Search



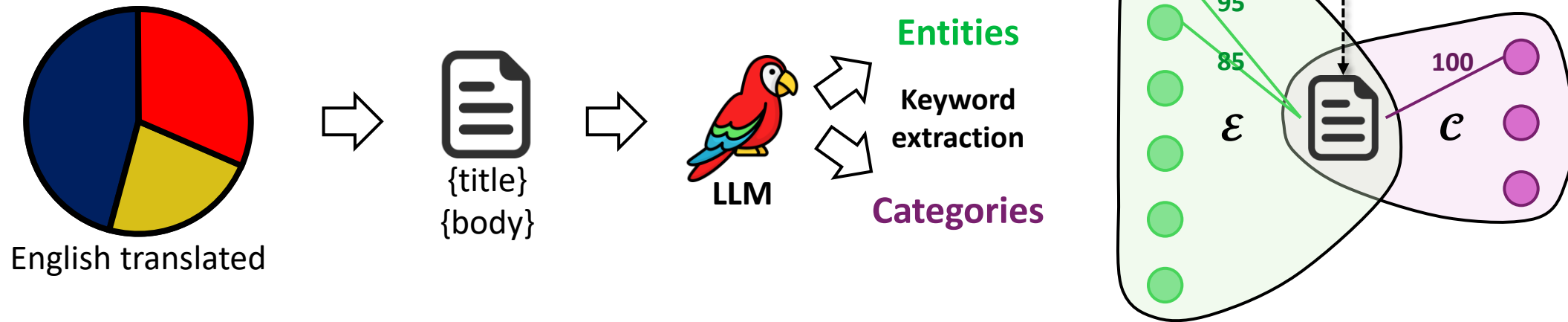
(3) Retrieval



Top K

DOCUMENT RETRIEVAL (WSDM CUP 26)

■ Our Approach : Tri-Partite Hyper Graph



Example

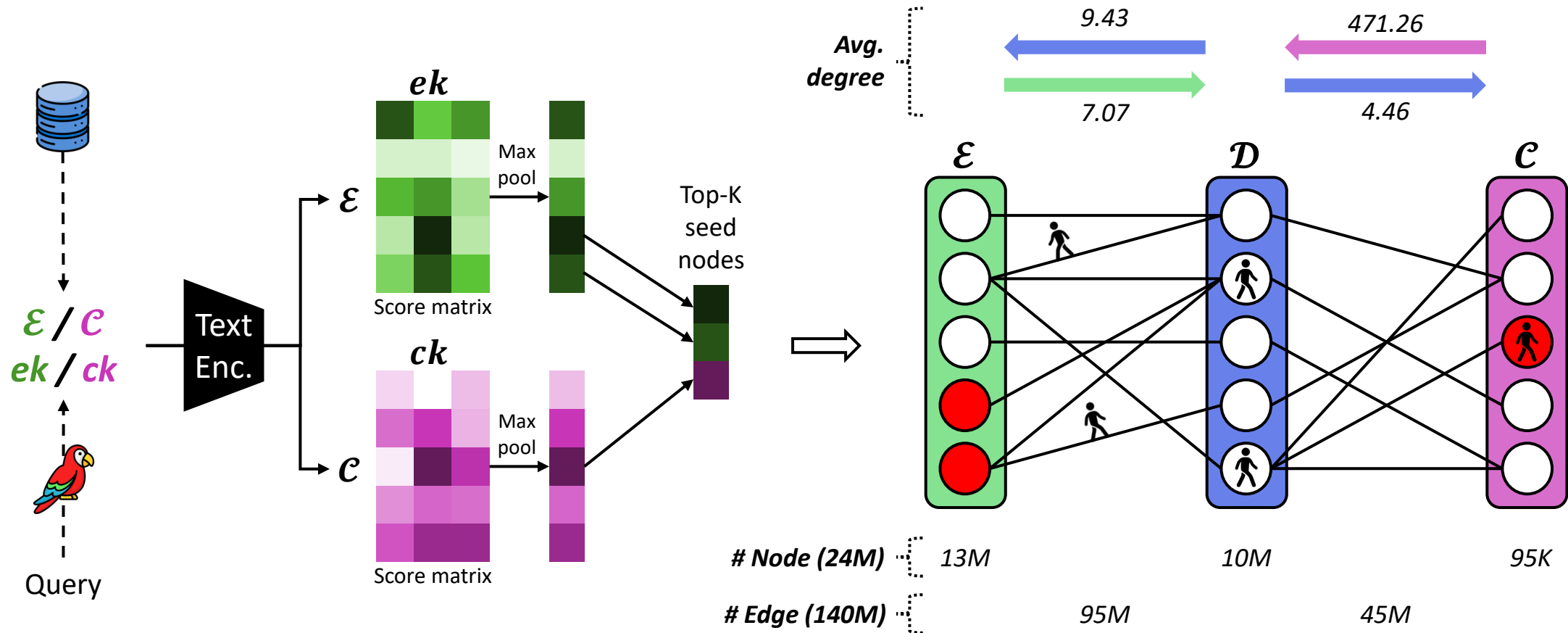
Oak cloud chain master heavy
online, together with "on the
chain Bloomberg" to listen to
the pulse of the industry
...

oculus cloud chain master (95)
Oklink (85)
Chainhub (85)
...
blockchain (100)
Technology (90)
...

{doc_id} {e1} {score}
{doc_id} {e2} {score}
{doc_id} {e3} {score}
...
{doc_id} {c1} {score}
{doc_id} {c2} {score}
...

DOCUMENT RETRIEVAL (WSDM CUP 26)

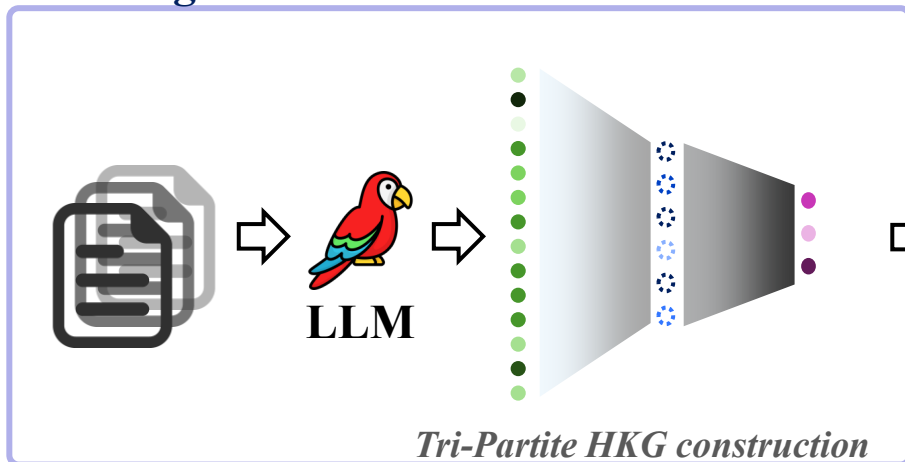
Retrieval Pipeline With Random Walk



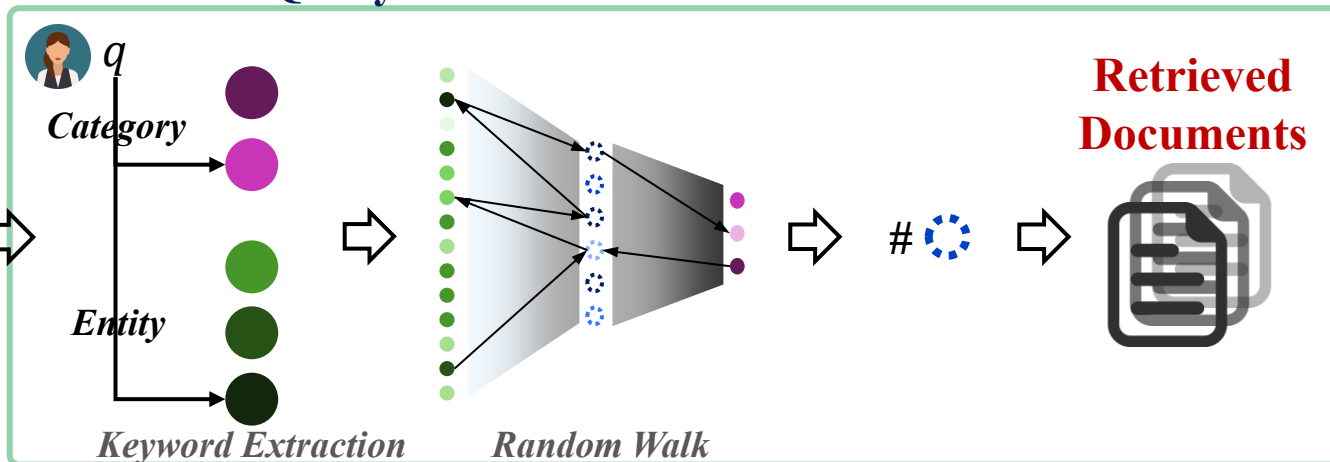
DOCUMENT RETRIEVAL (WSDM CUP 26)

Tri-Partite Retrieval via Random Walk (Tri-Walk)

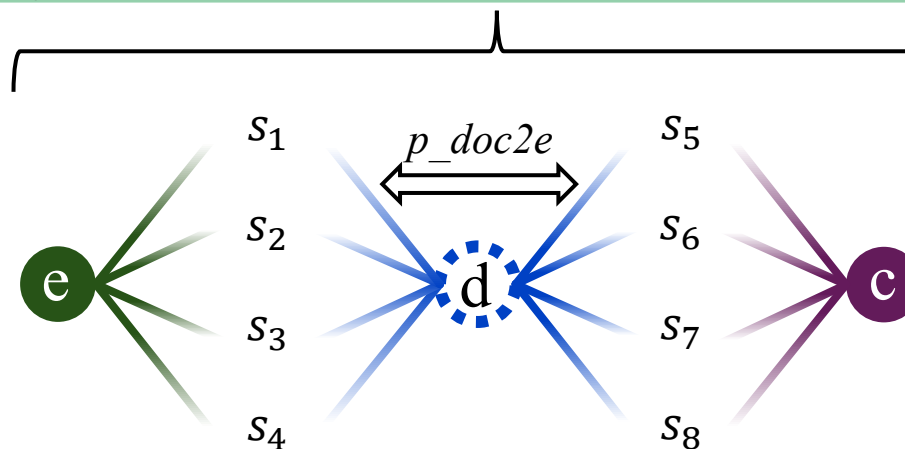
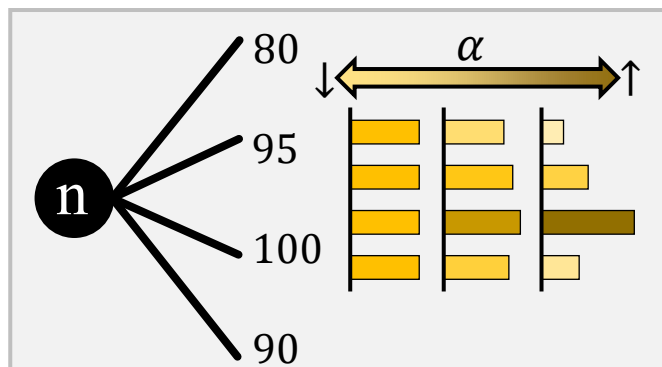
Indexing Time



Retrieval & Query Time

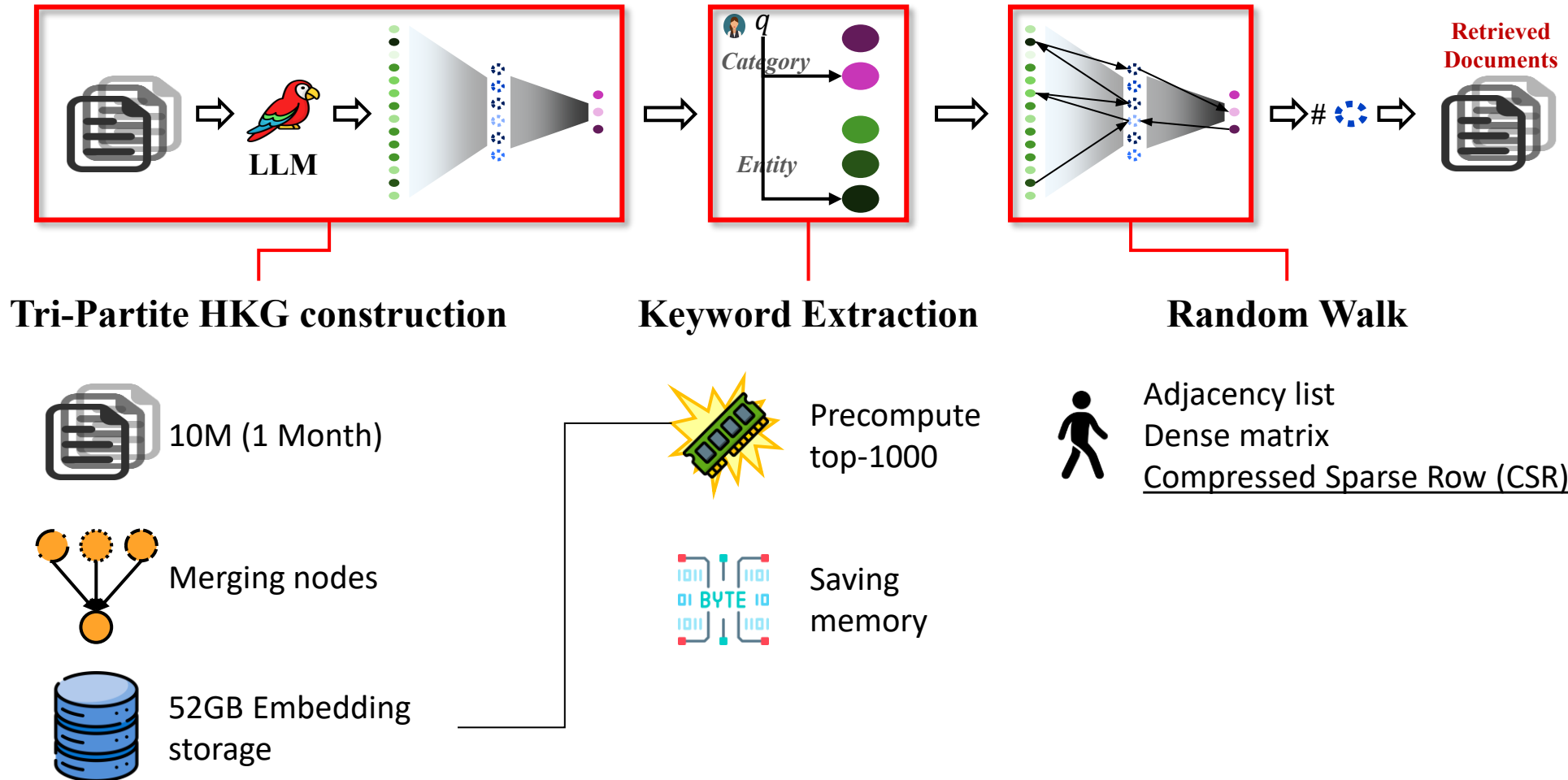


Probabilistic Transition



DOCUMENT RETRIEVAL (WSDM CUP 26)

■ Engineering Points (Memory, Time)



■ Evaluation on Development Queries

Query

“Iranian female athletes refugees
I am looking for stories about Iranian female
athletes who seek asylum in other countries.”

Documents

ab2e8958-2660-4d3a-ad3f-d631723d12df
1312ba58-66b4-41e6-8c80-e8fc80b556d2
0702fcd7-e85f-416d-b27d-99847fcfe07c
...

Relevance Score

3
1
1
...

Experiments

Hyperparameter
Grid Search

Types of walk decay

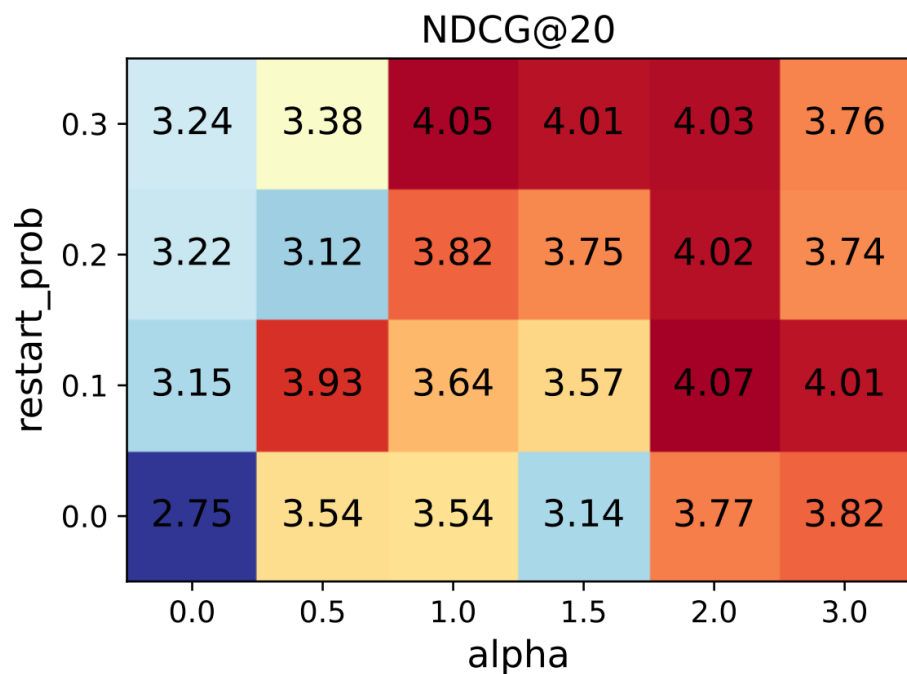
Metric

$$NDCG@k = \frac{DCG@k}{IDCG@k}$$

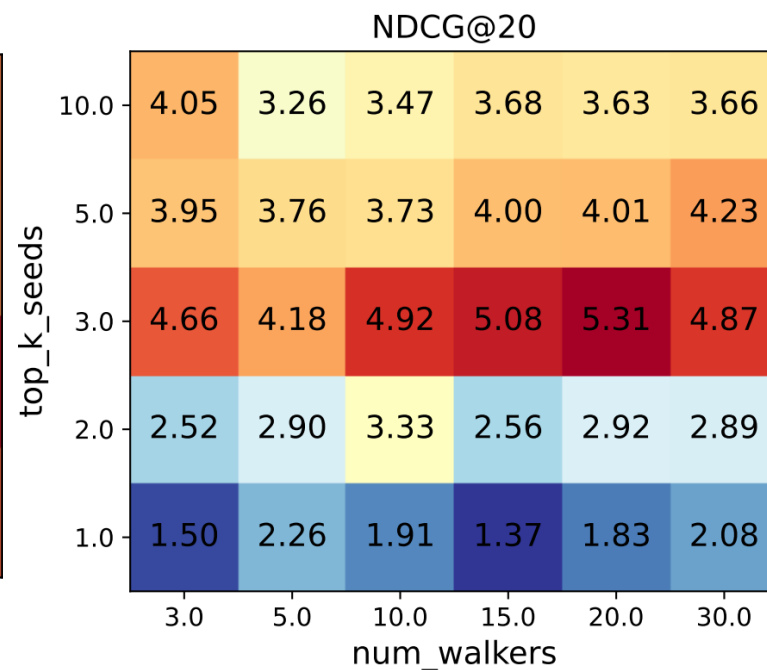
$$DCG@k = \sum_{i=1}^k \frac{2^{rel_i} - 1}{\log_2(i + 1)} \quad IDCG@k = \max(DCG@k)$$

EXPERIMENTS & DISCUSSION

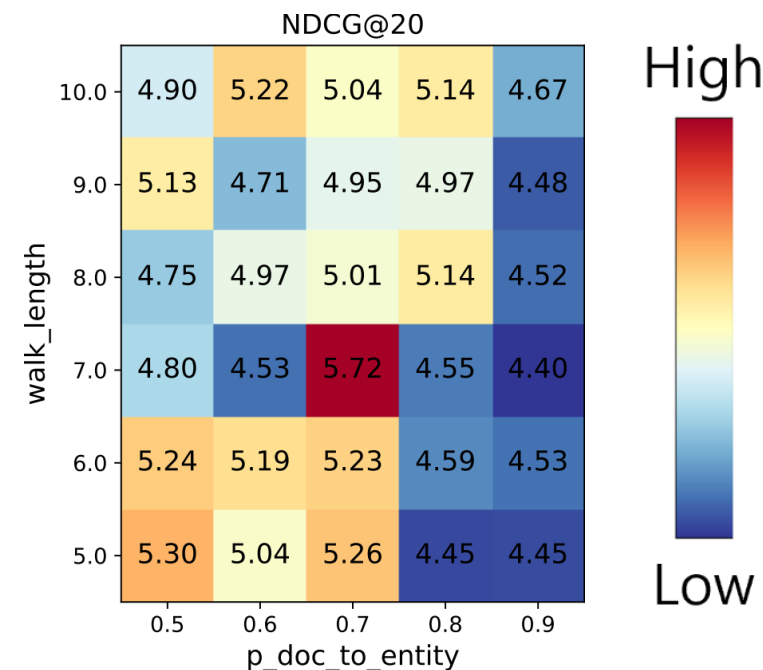
Hyperparameter Grid Search with NDCG@20 (%)



Restart prob : Less of an impact
Alpha : Favors sharp distribution



Top k seeds : High concentration required
Num walkers : High enough

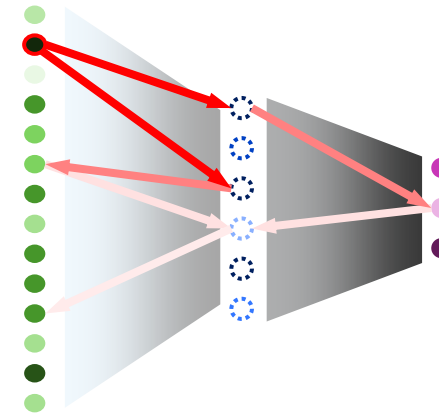
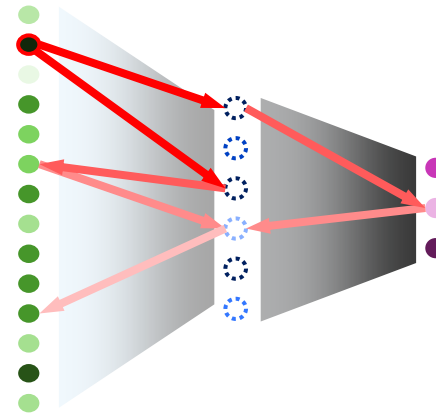
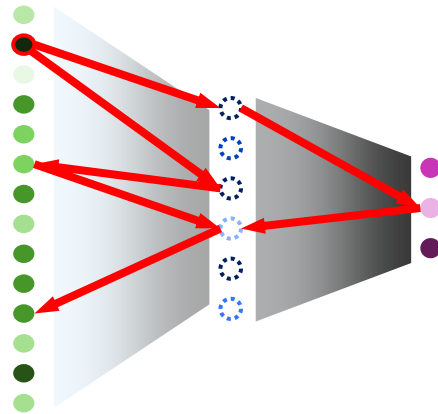


Walk length : Not too far
Prob D2E : Favors the entity side

EXPERIMENTS & DISCUSSION

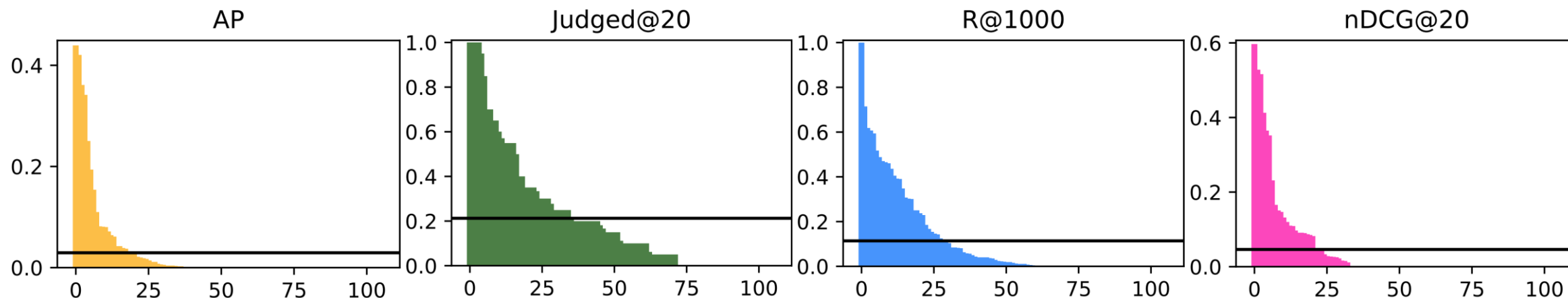
■ Walk Decay Types

$\text{score}[\text{doc}] += 1$	$\text{score}[\text{doc}] += 1/\sqrt{\text{step}}$	$\text{score}[\text{doc}] += 1/\text{step}$
5.72	<u>5.01</u>	4.50



EXPERIMENTS & DISCUSSION

■ WSDM CUP Results (105 Queries)



AP (%)	Judged@20 (%)	Recall@1000 (%)	NDCG@20 (%)
2.9	21.2	11.4	4.6

$$AP = \sum_{i=1}^k \frac{1[rel_i]}{i}$$

$$Judged@k = \frac{1}{k} \sum_{i=1}^k 1[rel_i]$$

$$Recall@k = \frac{1}{|\mathcal{R}|} \sum_{i=1}^k 1[rel_i]$$

$$NDCG@k = \frac{DCG@k}{IDCG@k}$$

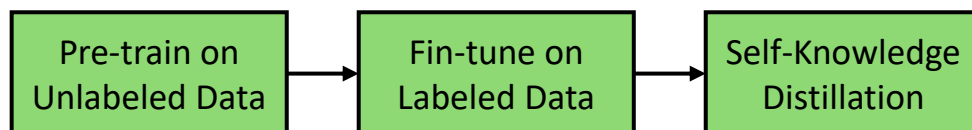
EXPERIMENTS & DISCUSSION

■ Baseline Methods & Our Mistakes

	Judged@20 (%)	NDCG@20 (%)
BGE-M3 Sparse	30.0	5.4
Rank-K (QwQ)	99.0	50.6
Ours	21.2	4.6

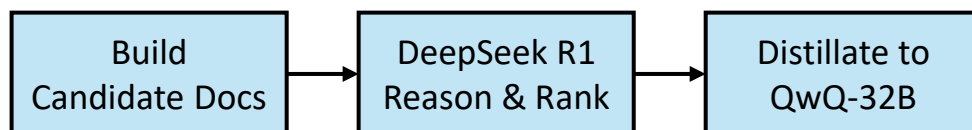
BGE-M3 Sparse

$32 \times A100 + 96 \times A800 = 8,960 \text{ GB}$



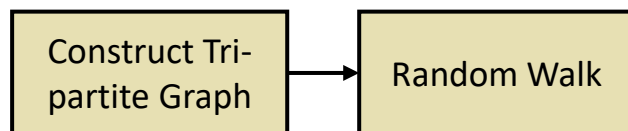
Rank-K (QwQ)

$128 \times \text{AMD MI250X} = 16,384 \text{ GB}$



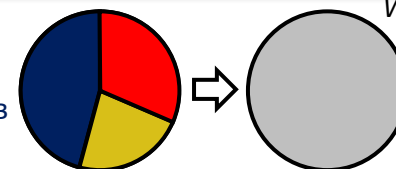
Tri-Walk

$1 \times A5000 + 2 \times \text{RTX4080} = 56 \text{ GB}$



Monolingual Loss

国内招聘情况不佳
Низкий уровень найма
отечественных работников
تسا فيعض يلخاد مادختسا

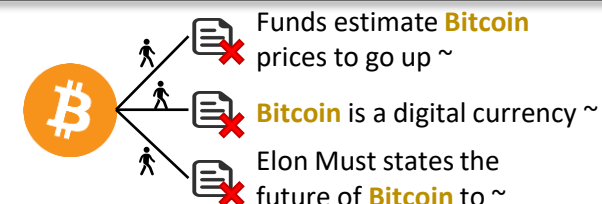


Which Country?

Domestic hiring is poor

Lost Concurrency

Articles on Bitcoin market crash



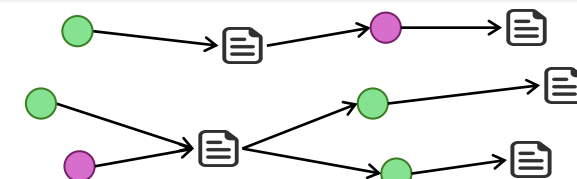
Funds estimate Bitcoin prices to go up ~

Bitcoin is a digital currency ~

Elon Must states the future of Bitcoin to ~

Path Semantics

What does 'walk' really give us?



CONCLUSION

■ From a High-Level View

- Referencing knowledge from outer source
- Reduce forgetting & avoid costly training

■ Graph RAG for LLMs

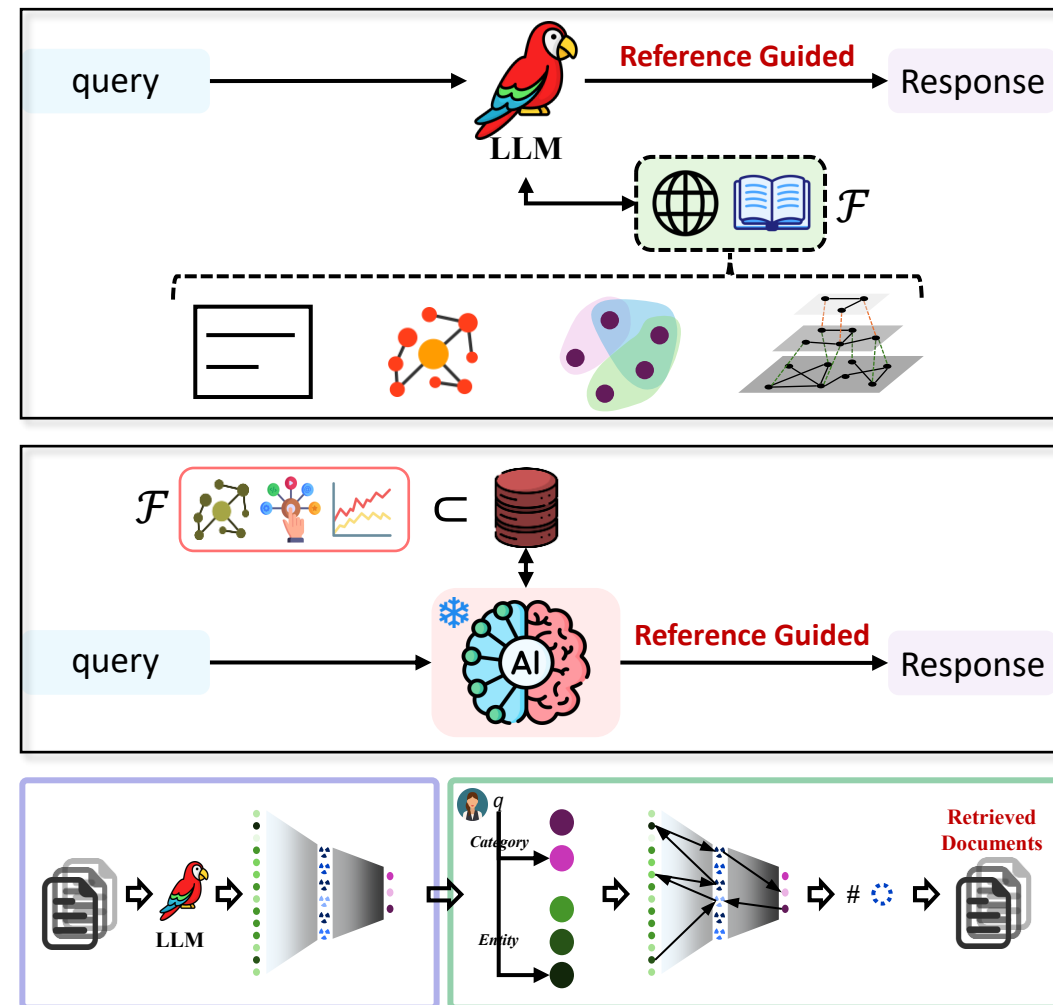
- Connecting dispersed information via graph

■ Retrieval Augmented Works

- Reference guided answering for various tasks
- Reference pool, retrieval module, response fusion

■ Practice with WSDM CUP 2026

- Multilingual document retrieval
- Handling practical issues for large scale retrieval



Q&A

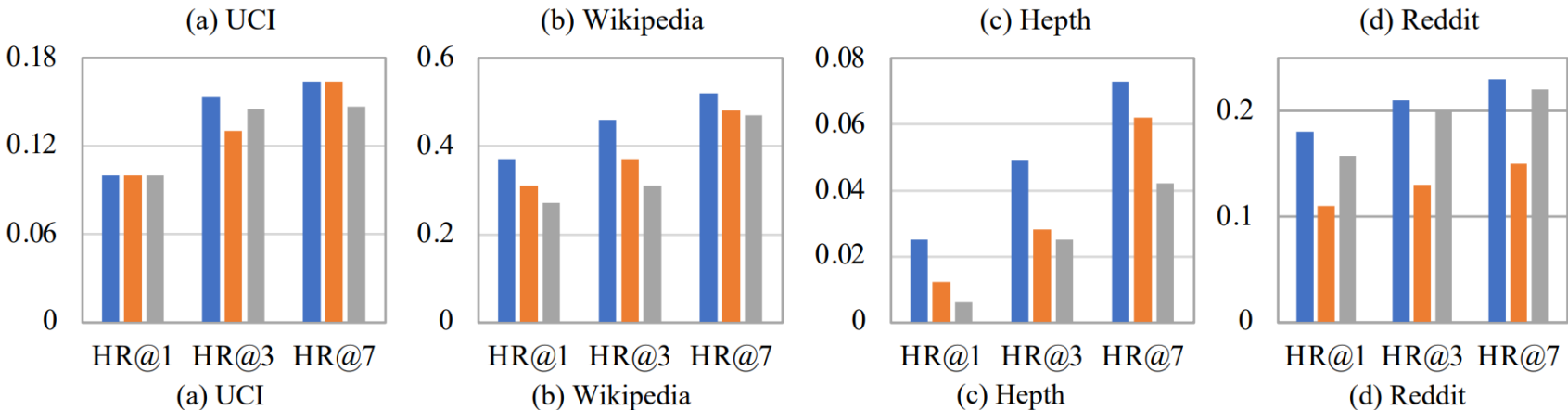
■ T1 : Experiment Results

Datasets	Models	DySAT	EvolveGCN	DyRep	JODIE	TGAT	TGN	TREND	GraphMixer	IDOL	SimpleDyG	RAG4DyG
UCI	Recall@5	0.009±0.003	0.072±0.046	0.009±0.008	0.018±0.019	0.022±0.004	0.014±0.010	0.083±0.015	0.097±0.019	0.093±0.029	<u>0.109</u> ±0.014	0.111 ±0.013
	NDCG@5	0.010±0.003	0.064±0.045	0.011±0.018	0.022±0.023	0.061±0.007	0.041±0.017	0.067±0.010	<u>0.104</u> ±0.013	0.075±0.022	<u>0.104</u> ±0.010	0.122* ±0.014
	Jaccard	0.010±0.001	0.032±0.026	0.010±0.005	0.012±0.009	0.020±0.002	0.011±0.003	0.039±0.020	0.042±0.005	0.014±0.002	<u>0.092</u> ±0.014	0.097 ±0.010
Hepth	Recall@5	0.008±0.004	0.008±0.002	0.009±0.006	0.010±0.008	0.011±0.007	0.011±0.006	0.010±0.008	0.009±0.002	0.007±0.002	<u>0.013</u> ±0.006	0.019* ±0.002
	NDCG@5	0.007±0.002	0.009±0.004	0.031±0.024	0.031±0.021	0.034±0.023	0.030±0.012	0.031±0.003	0.011±0.008	0.011±0.003	<u>0.035</u> ±0.014	0.045* ±0.003
	Jaccard	0.005±0.001	0.007±0.002	0.010±0.006	0.011±0.008	0.011±0.006	0.008±0.001	0.010±0.002	0.010±0.003	0.006±0.001	<u>0.013</u> ±0.006	0.019* ±0.002
MMConv	Recall@5	0.108±0.089	0.050±0.015	0.156±0.054	0.052±0.039	0.118±0.004	0.085±0.050	0.134±0.030	0.206 ±0.001	0.169±0.006	0.170±0.010	<u>0.194</u> ±0.005
	NDCG@5	0.102±0.085	0.051±0.021	0.140±0.057	0.041±0.016	0.089±0.033	0.096±0.068	0.116±0.020	0.172±0.029	0.115±0.039	<u>0.184</u> ±0.012	0.208* ±0.005
	Jaccard	0.095±0.080	0.032±0.017	0.067±0.025	0.032±0.022	0.058±0.021	0.066±0.038	0.060±0.018	0.085±0.016	0.015±0.002	<u>0.169</u> ±0.010	0.194* ±0.005
Wikipedia	Recall@5	0.003±0.005	0.012±0.01	0.003±0.002	0.017±0.005	0.006±0.004	0.016±0.018	0.022±0.012	0.010±0.007	0.022±0.008	<u>0.356</u> ±0.016	0.369* ±0.006
	NDCG@5	0.002±0.003	0.008±0.007	0.002±0.002	0.015±0.003	0.005±0.005	0.015±0.022	0.016±0.018	0.007±0.006	0.015±0.005	0.398 ±0.03	<u>0.389</u> ±0.008
	Jaccard	0.001±0.001	0.004±0.004	0.001±0.001	0.007±0.002	0.002±0.002	0.007±0.009	0.007±0.021	0.004±0.002	0.004±0.001	<u>0.320</u> ±0.027	0.328 ±0.007
Enron	Recall@5	0.002±0.004	0.004±0.011	0.021±0.001	0.005±0.005	0.020±0.002	0.001±0.001	0.023±0.003	0.021±0.002	0.024±0.014	<u>0.094</u> ±0.005	0.100* ±0.003
	NDCG@5	0.001±0.002	0.007±0.020	0.036±0.002	0.061±0.039	0.036±0.001	0.003±0.001	0.027±0.001	0.037±0.001	0.025±0.011	<u>0.114</u> ±0.005	0.119* ±0.004
	Jaccard	0.001±0.001	0.003±0.009	0.019±0.001	0.011±0.007	0.020±0.001	0.001±0.001	0.012±0.001	0.020±0.002	0.008±0.003	<u>0.068</u> ±0.003	0.071* ±0.002
Reddit	Recall@5	0.001±0.002	0.006±0.002	0.019±0.004	0.013±0.003	0.001±0.001	0.001±0.001	0.002±0.003	0.001±0.001	0.003±0.002	<u>0.101</u> ±0.019	0.119* ±0.006
	NDCG@5	0.001±0.002	0.012±0.003	0.020±0.004	0.015±0.002	0.001±0.001	0.002±0.001	0.003±0.002	0.003±0.001	0.005±0.003	<u>0.134</u> ±0.012	0.143 ±0.005
	Jaccard	0.001±0.001	0.003±0.001	0.013±0.004	0.007±0.002	0.001±0.001	0.001±0.002	0.001±0.001	0.002±0.001	0.002±0.001	<u>0.088</u> ±0.012	0.096 ±0.003

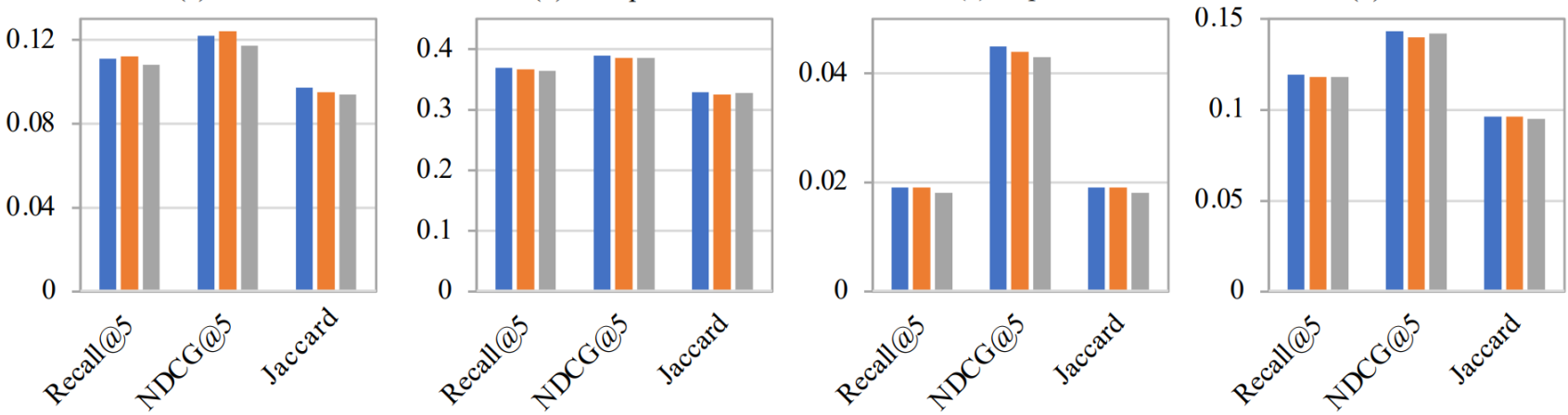
■ T1 : Impact of Context-aware Contrastive Learning and Decay

■ RAG4DyG ■ w/o CCL ■ w/o Decay

Retrieval
precision



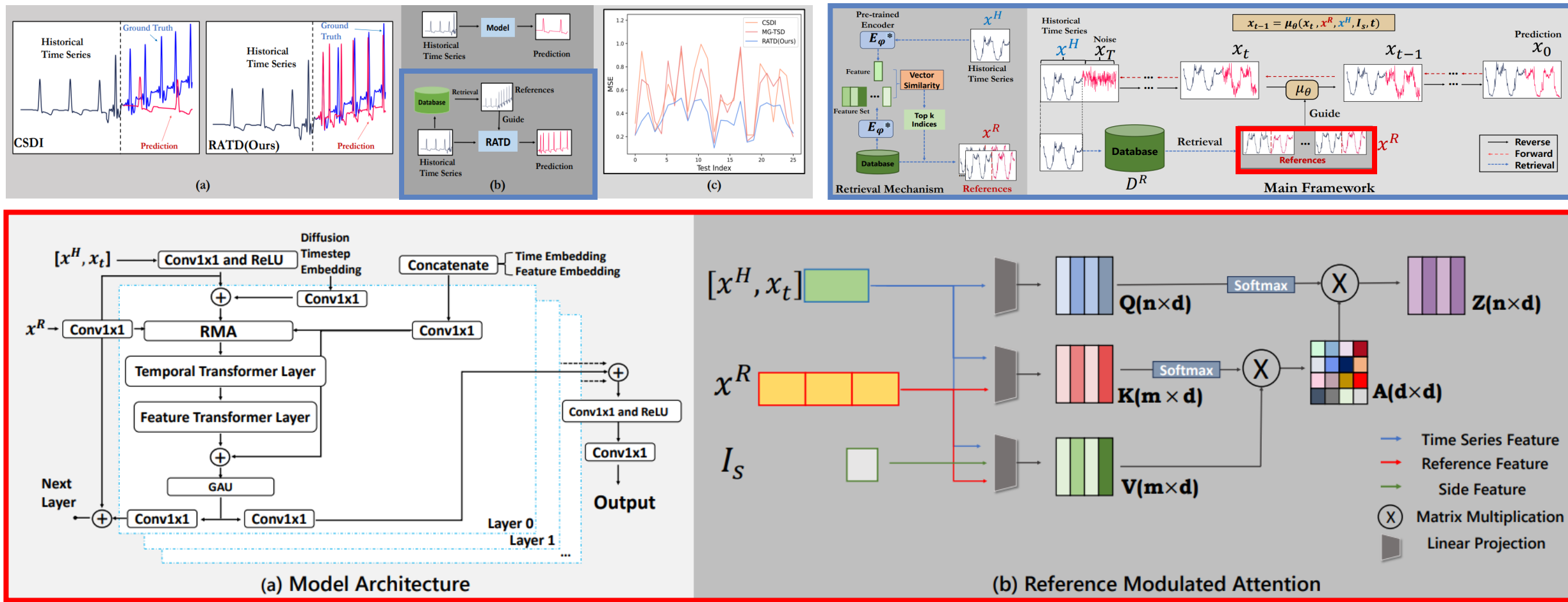
Link
prediction



■ T2 : Experiment Results (@20)

Method	TAOBAO		KOUBEI		AMAZON	
	Recall	nDCG	Recall	nDCG	Recall	nDCG
LightGCN	22.47±02.53	21.89±02.80	30.21±06.45	22.24±05.83	15.07±06.48	06.53±02.66
SGL	22.15±02.20	22.12±03.09	32.61±04.27	22.36±04.82	15.78±07.12	07.90±02.49
MixGCF	22.84±02.15	23.05±03.87	32.06±04.20	22.49±06.91	15.24±08.98	07.40±03.44
SimGCL	22.18±02.22	23.15±02.75	33.07±05.28	23.08±05.55	16.10±07.91	07.58±03.51
GraphPrompt	20.76±01.54	20.22±00.98	33.24±04.98	24.12±09.20	16.20±08.58	07.89±04.12
GPF	22.46±01.66	22.12±01.16	33.70±06.44	24.39±04.01	17.67±09.04	08.94±04.57
EvolveGCN-H	22.44±02.55	22.17±01.79	31.22±04.25	23.00±02.92	14.97±10.28	07.20±05.43
EvolveGCN-O	23.64±02.13	23.24±01.28	33.01±05.22	23.98±04.01	17.48±08.13	08.68±04.25
ROLAND	22.67±02.42	22.60±01.91	30.11±03.14	22.29±01.84	15.33±07.10	07.09±03.02
GraphPro+						
Vanilla/NF	20.10±01.50	20.12±01.30	21.31±04.59	15.31±03.11	12.56±07.45	06.31±03.92
Vanilla/FT	23.99±02.11	23.26±01.42	33.96±04.13	24.66±02.78	18.14±07.55	08.73±03.74
PRODIGY/NF	21.67±01.42	23.15±03.20	21.66±03.21	14.82±03.92	11.88±02.61	05.84±01.84
PRODIGY/FT	23.74±01.22	23.65±02.31	33.46±04.70	23.28±03.40	16.72±04.28	08.09±02.66
RAGRAPN/NF	20.31±01.60	20.45±01.44	22.86±03.44	16.68±02.48	13.78±05.54	06.52±02.69
RAGRAPN/FT	<u>24.78</u> ±01.93	<u>24.35</u> ±01.34	<u>34.27</u> ±03.93	<u>24.82</u> ±02.69	<u>18.69</u> ±07.45	<u>09.09</u> ±03.89
TarDGR/NF	20.39±02.41	20.91±02.18	24.83±03.68	17.90±02.62	14.26±05.37	06.74±02.56
TarDGR/FT	25.20 ±02.13	24.59 ±01.42	36.52 ±04.44	26.63 ±02.98	19.56 ±07.17	09.70 ±03.62

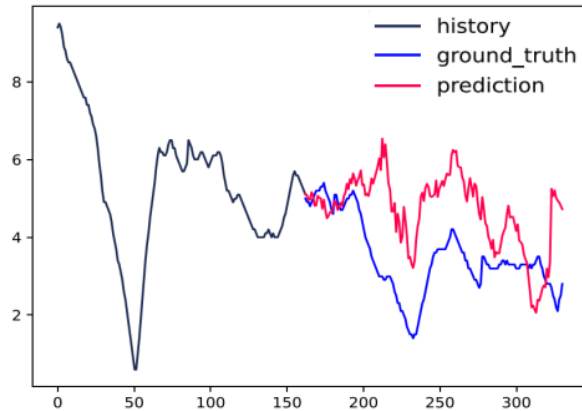
■ T3 : Aggregating Reference to Diffusion



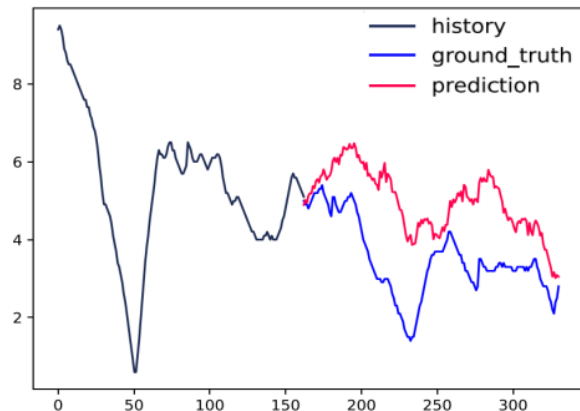
■ T3 : Experiment Results

Dataset	Exchange			Wind			Electricity			Weather		
Metric	MSE	MAE	CRPS	MSE	MAE	CRPS	MSE	MAE	CRPS	MSE	MAE	CRPS
RATD (ours)	0.013	0.073	0.339	0.784	0.579	0.673	<u>0.151</u>	<u>0.246</u>	0.373	0.281	0.293	0.301
TimeDiff	0.018	0.091	0.589	0.896	0.687	0.917	0.193	0.305	0.490	0.327	<u>0.312</u>	0.410
CSDI	0.077	0.194	0.397	1.066	0.741	0.941	0.379	0.579	0.480	0.356	0.374	0.354
mr-Diff	<u>0.016</u>	0.082	0.397	<u>0.881</u>	0.675	0.881	0.173	0.258	0.429	0.296	0.324	0.347
D ₃ VAE	0.200	0.301	0.401	1.118	0.779	0.979	0.286	0.372	<u>0.389</u>	0.315	0.380	0.381
Fedformer	0.133	0.233	0.631	1.113	0.762	1.235	0.238	0.341	0.561	0.342	0.347	0.319
FreTS	0.039	0.140	0.440	1.004	0.703	0.943	0.269	0.371	0.634	0.351	0.354	0.391
FiLM	<u>0.016</u>	0.079	0.349	0.984	0.717	0.798	0.210	0.320	0.671	0.327	0.336	0.556
iTransformer	<u>0.016</u>	<u>0.074</u>	<u>0.343</u>	0.932	0.676	<u>0.811</u>	0.192	0.262	0.402	0.358	0.401	<u>0.318</u>
Autoformer	0.056	0.167	0.769	1.083	0.756	1.201	1.026	0.313	0.602	0.360	0.354	0.754
Pyraformer	0.032	0.112	0.532	1.061	0.735	0.994	0.273	0.379	0.732	0.394	0.385	0.485
Informer	0.073	0.192	0.631	1.168	0.772	1.065	0.292	0.383	0.749	0.385	0.364	0.821
PatchTST	0.047	0.153	0.629	1.001	<u>0.672</u>	1.026	0.225	0.394	0.801	0.782	0.670	0.370
SCINet	0.038	0.137	0.624	1.055	0.732	0.997	0.171	0.280	0.499	0.329	0.344	0.814
DLinear	0.022	0.102	0.538	0.899	0.686	0.957	0.215	0.336	0.527	0.488	0.444	0.791
NLinear	0.019	0.091	0.481	0.989	0.706	0.974	0.147	0.239	0.419	0.369	0.328	0.738
TimesNet	0.023	0.120	0.520	0.982	0.771	1.001	0.141	0.361	0.403	<u>0.313</u>	0.364	0.491
NBeats	<u>0.016</u>	0.081	0.399	1.069	0.741	0.981	0.269	0.370	0.697	<u>0.744</u>	0.420	0.871

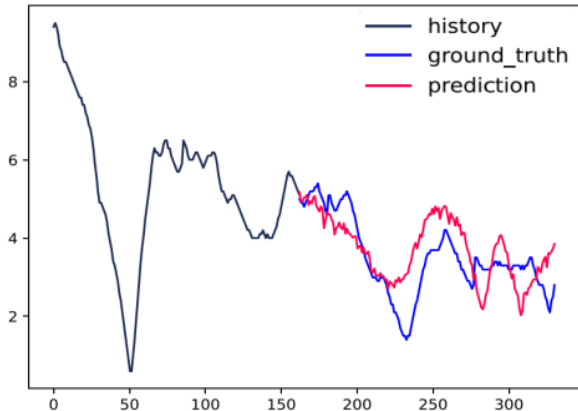
■ T3 : Case Study



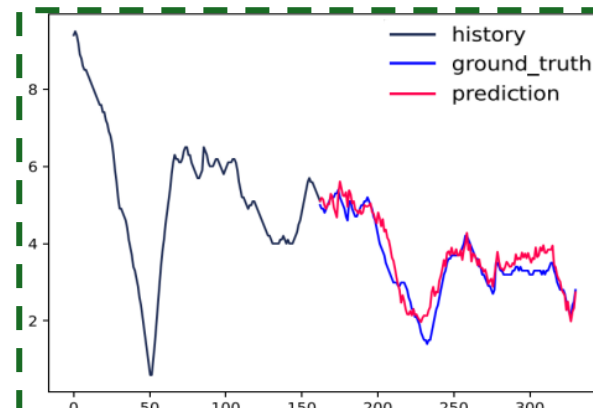
CSDI



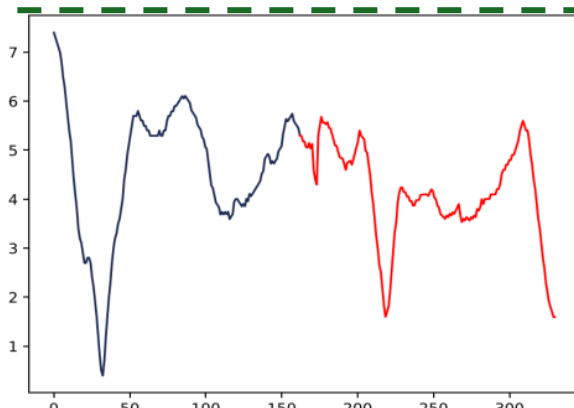
D³ VAE



iTransformer



RATD



Reference

■ T4 : Experiment Results

		Test-I				Test-II				Test-ALL			
		Hits@N (%)			MRR	Hits@N (%)			MRR	Hits@N (%)			MRR
		3	5	10		3	5	10		3	5	10	
With Rules	KALE-Pre (Guo et al., 2016)	35.8	41.9	49.8	0.291	82.9	86.1	89.9	0.713	61.7	66.2	71.8	0.523
	KALE-Joint (Guo et al., 2016)	38.4	44.7	52.2	0.325	79.7	84.1	89.6	0.684	61.2	66.4	72.8	0.523
	ASR-DistMult (Minervini et al., 2017)	36.3	40.3	44.9	0.330	98.0	99.0	99.2	0.948	70.7	73.1	75.2	0.675
	ASR-ComplEx (Minervini et al., 2017)	37.3	41.0	45.9	0.338	99.2	99.3	99.4	0.984	71.7	73.6	75.7	0.698
	KBLR (Garcia-Duran and Niepert, 2018)	–	–	–	–	–	–	–	–	74.0	77.0	79.7	0.702
Without Rules	TransE (Bordes et al., 2013)	36.0	41.5	48.1	0.296	77.5	82.8	88.4	0.630	58.9	64.2	70.2	0.480
	DistMult (Yang et al., 2015)	36.0	40.3	45.3	0.313	92.3	93.8	94.7	0.874	67.4	70.2	72.9	0.628
	ComplEx (Trouillon et al., 2016)	37.0	41.3	46.2	0.329	91.4	91.9	92.4	0.887	67.3	69.5	71.9	0.641
	GNTPs (Minervini et al., 2020)	33.7	36.9	41.2	0.313	98.2	99.0	99.3	0.977	69.2	71.1	73.2	0.678
	RotatE (Sun et al., 2019)	51.1	55.1	60.3	0.471	86.8	88.6	90.7	0.846	70.8	73.57	77.0	0.678
	CBR (Das et al., 2020)	40.0	44.5	48.8	0.359	67.8	71.8	75.9	0.636	57.0	61.2	65.3	0.527
	Our Model	49.0	52.7	57.1	0.457	94.8	95.0	95.3	0.948	74.2	76.0	78.2	0.727

■ T4 : Experiment Results Under The Open-World

